14 Pulp Reactions to Caries and Dental Procedures

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The dental pulp is a dynamic tissue that responds to external stimuli in a variety of ways. However, there are certain unique features of the dental pulp response that distinguish it from other connective tissues in the body. These include the pulp's exposure to dental caries, a very prevalent chronic infectious disease, its encasement in an unyielding environment after complete tooth maturation, and the paucity of collateral circulation render it susceptible to injury and complicate its regeneration. Moreover, the pulp is endowed with a rich neurovascular supply that promotes the effects of inflammation and may lead to rapid degeneration and necrosis. The treatment of dental caries and other tooth pathosis often involves the cleaning and shaping of the enamel and dentin, the hardest tissues in the body, thus adding to the irritation of the pulp. In this chapter, the response of the pulp to all these variables will be discussed and recent advances in our understanding of dental procedures and their effects on the pulp will be presented.

Pulp Reaction to Caries

Dental caries is a localized, destructive, and progressive infection of dentin, which, if left unchecked, can result in pulp necrosis and potential tooth loss. Both bacterial by-products and products from the dissolution of the organic and inorganic constituents of dentin mediate the effects of dental caries on the pulp. Three basic reactions tend to protect the pulp against caries: (1) decreases in dentin permeability due to dentin sclerosis, (2) tertiary dentin formation, and (3) inflammatory and immune reactions in the pulp.[143] These responses occur concomitantly and their robustness is highly dependent on the aggressive nature of the advancing lesion.

Bacterial proteolytic enzymes, toxins, and metabolic by-products have been thought to be initiators of pulp reactions, yet the buffering capacity of dentin and dentinal fluid likely attenuate these stimulatory effects. This protective function is significantly reduced when the remaining dentin thickness (RDT) is minimal.[267] When the pulp is exposed, bacterial metabolites, toxins, and cell wall components induce inflammation. In initial to moderate lesions, current evidence suggests that acidic by-products of the carious process act indirectly by degrading the dentin matrix, thereby liberating bioactive molecules previously sequestered during dentinogenesis. Once liberated, these molecules resume their role in dentin formation, this time being stimulatory for *tertiary* dentinogenesis.[268] This theory is supported by the findings that demineralized dentin matrix implanted at the site of pulp exposure can induce dentinogenesis.[300] Furthermore, placement of purified dentin matrix proteins on exposed dentin or exposed pulp stimulates tertiary dentin formation, indicating that these molecules can act directly or across intact dentin.[269,299]

Recent evidence suggests several candidate molecules for stimulation of reparative dentinogenesis. Transforming growth factor-β1 (TGF-β1), TGF-β3, insulin-like growth factors I (IGF-I) and II, platelet-derived growth factor (PDGF), bone morphogenetic protein-2 (BMP-2), and angiogenic growth factors such as vascular endothelial growth factor (VEGF) have been shown to be stimulatory for dentinogenesis in vitro.[259] The TGF-β superfamily in particular seems to be important in the signaling process for odontoblast differentiation as well as primary and tertiary dentinogenesis. As the predominant isoform, TGF-β1 is equally distributed in the soluble and insoluble fractions of dentin matrix.[43] During the carious dissolution of dentin, it is believed that the soluble pool of TGF-β1 can diffuse across intact dentin while the insoluble pool is immobilized on insoluble dentin matrix and serves to stimulate odontoblasts, much like membrane-bound TGF-βs during odontogenesis.[267,270]

Despite the research interest in tertiary dentinogenesis, it is neither the first nor necessarily the most effective defense against invading pathogens. A combination of an increased deposition of intratubular dentin and the direct deposition of mineral crystals into the narrowed dentin tubules to decrease dentin permeability is the first defense to caries and is called *dentin sclerosis*. The combination of increased deposition of intratubular dentin and tubule occlusion by precipitated crystals results in an effective decrease in dentin permeability underneath the advancing carious lesion.[229] In vitro studies with cultured tooth slices implicate TGF-β1 as a central player in the increased deposition of intratubular dentin.[266] The deposition of whitlockite crystals in the tubular lumen most likely results from a similar stimulation of vital associated odontoblasts, possibly in combination with precipitation of mineral released during the demineralization process (Fig. 14.1).[171,296]

The formation of tertiary dentin occurs over a longer period than does that of sclerotic dentin, and its resultant character is highly dependent on the stimulus. Mild stimuli activate resident quiescent odontoblasts whereupon they elaborate the organic matrix of dentin. This type of tertiary dentin is referred to as *reactionary dentin* and can be observed when initial dentin demineralization occurs beneath the noncavitated enamel lesion.[164] Mediators present during the carious process induce a focal upregulation of matrix production by resident odontoblasts. The resultant dentin is similar in morphology to physiologic dentin and may only be apparent due to a change in the direction of the new dentinal tubules (Fig. 14.2). In the aggressive lesion, the carious process may prove cytocidal to subjacent odontoblasts and require repopulation of the disrupted odontoblast layer with differentiating progenitors. The organization and composition of the resultant matrix are a direct reflection of the differentiation state of the secretory cells. This accounts for the heterogeneity of *reparative dentin*, where the morphology can range from organized tubular dentin to more disorganized irregular *fibrodentin*. Fibrodentin, due to its irregular configuration and tissue inclusions, is more permeable than physiologic dentin (Fig. 14.3).[310]

While dentin can provide a physical barrier against noxious stimuli, the pulp immune response provides humoral and cellular challenges to invading pathogens. In the



Fig. 14.2 Reactionary dentinogenesis *(TD)*. Note the tubular morphology and the discontinuity of the tubules at the interface of secondary and reactionary dentin. Resident odontoblasts are still present.

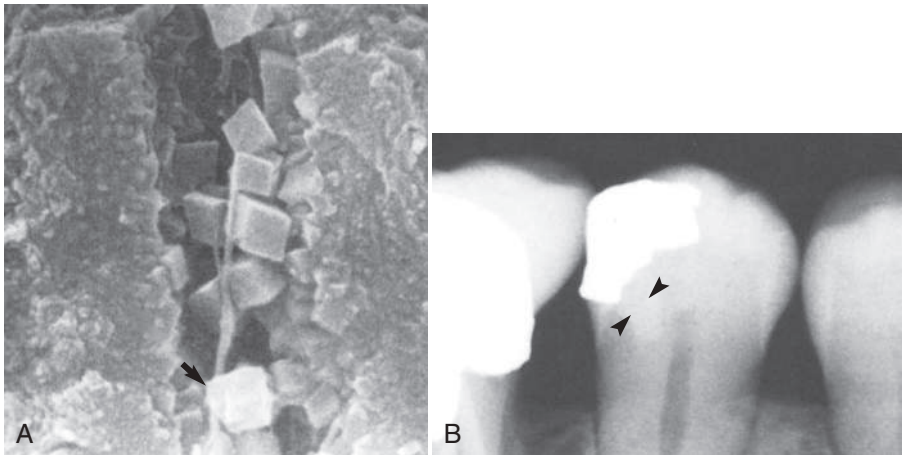


Fig. 14.1 **A,** Whitlockite crystals occlude the dentinal tubules in sclerotic dentin. **B,** Dentinal sclerosis is radiographically apparent beneath a deep class II lesion.

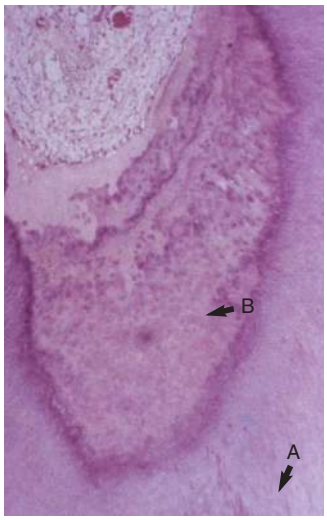


Fig. 14.3 Tertiary dentin. The strong stimulus of the impinging infection is cytocidal for odontoblasts. The resultant dentin is irregular with soft-tissue inclusions. Intact secondary dentin is depicted by *arrow A*, and tertiary dentin is designated by *arrow B*.

progressing carious lesion, the host immune response increases in intensity as the infection advances. It has been shown that titers of T helper cells, B-lineage cells, neutrophils, and macrophages are directly proportional to lesion depth in human teeth.[126] The disintegration of large amounts of dentin, however, is not necessary to elicit a pulp immune response. This is supported by the observation that a pulp inflammatory response can be seen beneath noncavitated lesions and noncoalesced pits and fissures.[36]

The early inflammatory response to caries is characterized by the focal accumulation of chronic inflammatory cells (Fig. 14.4). This is mediated initially by odontoblasts and later by dendritic cells. As the most peripheral cell in the pulp, the odontoblast is positioned to encounter foreign antigens first and initiate the innate immune response. Pathogen detection in general is accomplished via specific receptors called pattern recognition receptors (PRRs).[128] These receptors recognize pathogen associated molecular patterns (PAMPs) on invading organisms and initiate a host

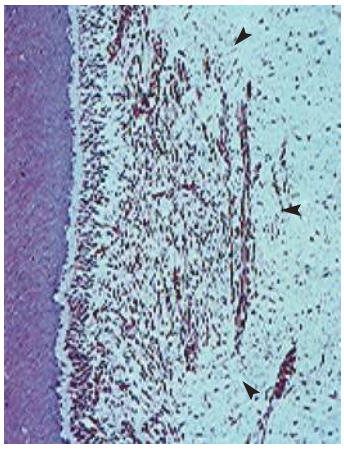


Fig. 14.4 The early pulpal response to caries is represented by a focal accumulation of chronic inflammatory cells *(arrowheads)*. Note that peripheral to the inflammation the pulpal parenchyma is relatively unaffected.

defense through the activation of the NF-κB pathway.[106] One class of the PAMP recognition molecules is the Toll-like receptor (TLR) family. Odontoblasts have been shown to have an increased expression of certain TLRs in response to bacterial products. Under experimental conditions, odontoblast expression of TLR3, 5, and 9 was increased in response to lipoteichoic acid while lipopolysaccharide increased TLR2 and 4 expression.[71,114,204] It was also shown that TGF-β1 inhibits the expression of TLR2 and 4 by odontoblasts in response to gram-positive and -negative bacteria.[114] Once the odontoblast TLR is stimulated by a pathogen, proinflammatory cytokines, chemokines, and antimicrobial peptides are elaborated by the odontoblast resulting in recruitment and stimulation of immune effector cells as well as direct bacterial killing.[80,82,138]

Many cells produce chemokines at low levels constitutively. Unstimulated odontoblasts express genes coding for CCl2, CXCL12, and CXCL14, three genes known to code for factors chemotactic for immature dendritic cells.[44] They also produce CCL26—a natural antagonist for CCR1, CCR2, and CCR5—which are chemokines normally produced by monocytes and dendritic cells.[324] Stimulation with bacterial cell wall constituents has been shown to upregulate the expression of multiple chemokine genes, including CXCL12, CCL2, CXCL9, CX3CL1, CCL8, CXCL10, CCL16, CCL5, CXCL2, CCL4, CXCL11, and CCL3, as well as 9 chemokine receptor genes, including CXCR4, CCR1, CCR5, CX3CR1, CCR10, and CXCR3, suggesting that odontoblasts sense pathogens and express factors that recruit immune effector cells (Fig. 14.5).[44,81,113,165] These data suggest a scenario whereby stimulated odontoblasts express high levels of chemokines, such as IL-8 (CXCL8), that act in concert with the release of formerly sequestered growth factors from carious dentin, which induce a focal increase in dendritic cell numbers with additional release of chemotactic mediators.[83,270] The subsequent influx of immune effector cells is composed of lymphocytes, macrophages, and plasma cells. This cellular infiltrate is accompanied by localized capillary sprouting in response to angiogenic factors as well as coaggregation of nerve fibers and HLA-DR-positive dendritic cells.[316,317]

As the carious lesion progresses, the density of the chronic inflammatory infiltrate and well as that of dendritic cells in the odontoblast region increases. Pulp dendritic cells are responsible for antigen presentation and stimulation of T lymphocytes. In the uninflamed pulp, they are scattered throughout the pulp. With caries progression, they aggregate initially in the pulp and subodontoblastic regions then extend into the odontoblast layer and eventually migrate into the entrance to tubules beside the odontoblast process (Fig. 14.6).[318] There are two distinct populations of dendritic cells that have been identified in the dental pulp. CD11c+ is found in the pulp/dentin border and subjacent to pits and fissures. F4/80+ dendritic cells are concentrated in the perivascular spaces in the subodontoblastic zone and inner pulp.[324] CD11c+ dendritic cells express TLRs 2 and 4 and are CD205 positive. F4/80+ dendritic cells have migratory ability. As they migrate from the central pulp, they increase in size and become CD86 positive. In vitro studies have suggested that the secretion of GM-CSF and osteopontin by dendritic cells and macrophages represents a mechanism whereby they contribute to odontoblast differentiation.[255]

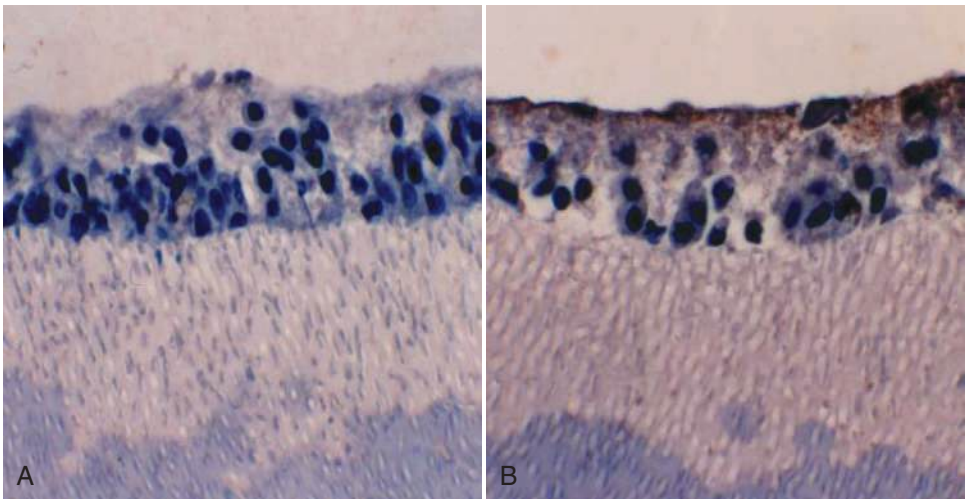


Fig. 14.5 Odontoblasts exposed to lipopolysaccharide (LPS) in an in vitro culture model express IL-8 as evidenced by immunostaining with anti-IL-8 antibodies. **A,** Control cultures not exposed to LPS (negative control). **B,** Immunostaining for IL-8 in cultures exposed to LPS.

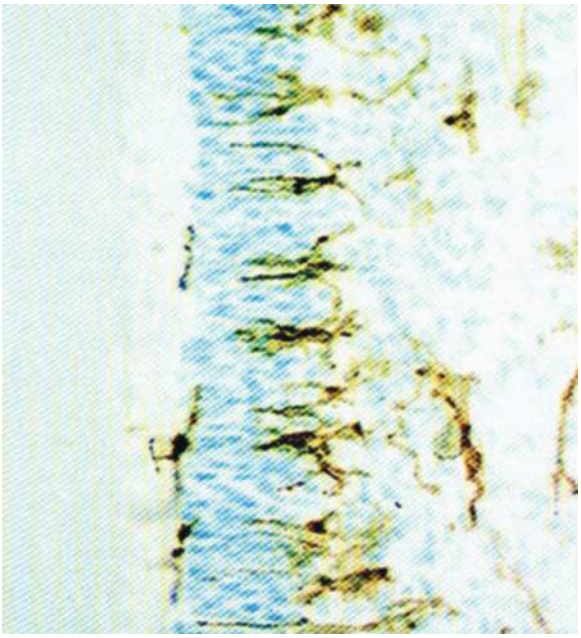


Fig. 14.6 **Dental caries stimulates the accumulation of pulpal dendritic cells in and around the odontoblastic layer.** (Reprinted with permission from Mats Jontell.)

Evidence suggests that odontoblasts also play a role in the humoral immune response to caries. IgG, IgM, and IgA have been localized in the cytoplasm and cell processes of odontoblasts in human carious dentin, suggesting that these cells actively transport antibodies to the infection front.[211] In the incipient lesion antibodies accumulate in the odontoblast layer and with lesion progression can be seen in the dentinal tubules. Eventually, this leads to a focal concentration of antibodies beneath the advancing lesion.[210]

In the most advanced phase of carious destruction, the humoral immune response is accompanied by immunopathologic destruction of pulp tissue. In animal studies where monkeys were hyperimmunized to BSA, there was an observed increase in pulp tissue destruction subsequent to antigenic challenge across freshly cut dentin.[22] Odontoblasts also appear to be involved in the production of innate antimicrobial molecules such as human beta defensin-2 (HBD2). IL-1 and TNF-α as well as bacterial LPS were responsible for significant increases in HBD2 in response to caries.[113] In sum, it appears that the odontoblasts play a central role in orchestrating local and chemotactic inflammatory responses to dental caries (Fig. 14.7).

Pulp exposure in primary and immature permanent teeth can lead to a proliferative response or *hyperplastic pulpitis*. Exuberant inflammatory tissue proliferates through the exposure and forms a "pulp polyp" (Fig. 14.8). It is presumed that a rich blood supply facilitates this proliferative response. Conventional root canal therapy or therapeutic pulpotomy would be indicated in these cases.

Neurogenic Mediators

Neurogenic mediators are involved in the pulp response to irritants and like immune components they can mediate pathology as well as the healing response. External stimulation of dentin causes the release of proinflammatory neuropeptides from pulp afferent nerves.[38,139] Substance P (SP), calcitonin gene-related peptide (CGRP), neurokinin A (NKA), neurokinin Y (NKY), and vasoactive intestinal peptide (VIP) are released and affect vascular events such as vasodilatation and increased vascular permeability. This results in a net increase in tissue pressure that can progress to necrosis in extreme and persistent circumstances. Stimulation of sympathetic nerves in response to the local release of mediators such as norepinephrine, neuropeptide Y, and ATP has been shown to alter pulp blood flow (PBF). Receptor field studies as well as anatomic studies have shown sprouting of afferent fibers in response to inflammation.[38]

Neuropeptides can act to modulate the pulp immune response. It has been demonstrated that SP acts as a chemotactic and stimulatory agent for macrophages and T lymphocytes. The result of this stimulation is increased production of arachidonic acid metabolites, stimulation of lymphocytic mitosis, and production of cytokines. CGRP demonstrates immunosuppressive activity, which

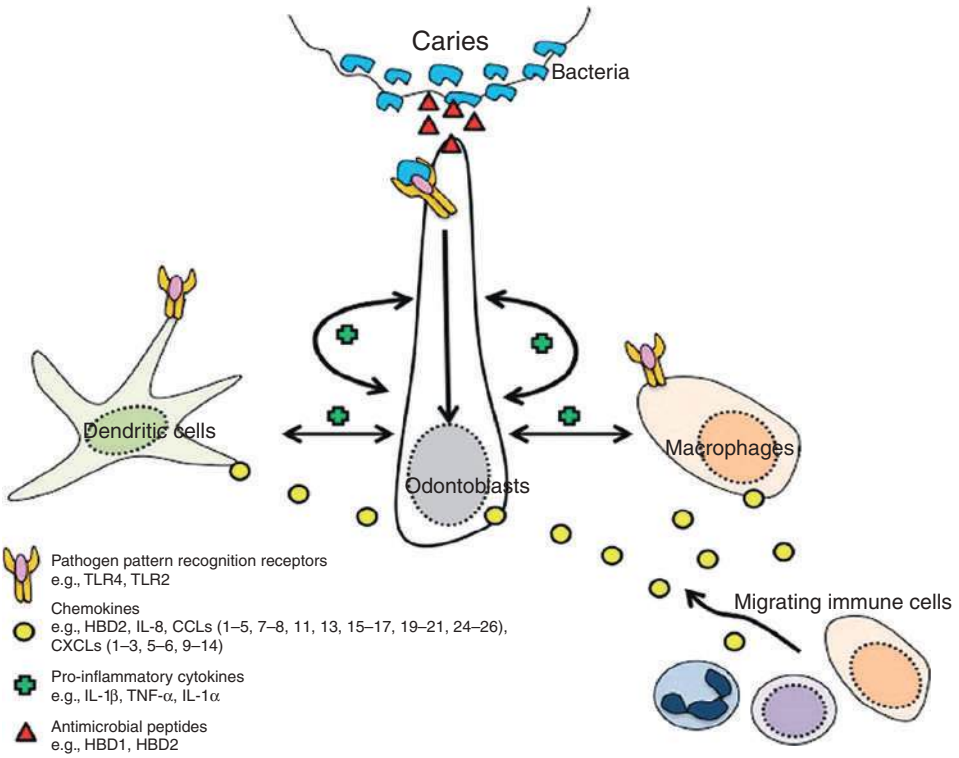


Fig. 14.7 Innate immunity in the odontoblast layer (ODL). Bacterial components from caries activate cytokine/chemokine release from odontoblasts, dendritic cells, and/or macrophages via Toll-like receptors *(TLRs)*. Proinflammatory cytokines released from these cells act as autocrine and paracrine signals to amplify cytokine responses including antimicrobial peptide, cytokine, and chemokine production. The release of chemokines creates a migration gradient for immune cells to ODL while antimicrobial peptides reduce bacterial load. (Reprinted with permission from Horst OV, Horst JA, Samudrala R, Dale BA. Caries induced cytokine network in the odontoblast layer of human teeth. *BMC Immunol* 12:9, 2011, Fig. 4C.)



Fig. 14.8 A proliferative response to caries in a young tooth, typically referred to as proliferative pulpitis, hyperplastic pulpitis, or pulp polyp. (Courtesy Dr. Howard Strassler at the University of Maryland, with permission.)

is evidenced by a diminution of Class II antigen presentation and lymphocyte proliferation.

SP and CGRP are mitogenic for pulp and odontoblast-like cells, thereby initiating and propagating the pulp healing response.[294] CGRP has been shown to stimulate the production of bone morphogenic protein by human pulp cells. The result of such stimulation has been postulated to induce tertiary dentinogenesis.[40] SP appears to increase in the dental pulp and periodontal ligament as a result of acutely induced occlusal trauma,[47] which may be related to the pain associated with concussion traumatic injury.

It was shown that there may be gender differences in CGRP production in the dental pulp.[31,175] In one study, serotonin (a peripheral pronociceptive mediator) induced a significant increase in capsaicin-evoked CGRP release in dental pulps obtained from female and not male patients.[175] This interplay of inflammatory mediators may explain some of the gender differences in clinical presentation with dental pain.

It was also recently shown that these neurotropic factors are not only present in dentin matrix but may be released and may aid in trigeminal neurite growth during regenerative procedures of the dental pulp.[13,313]

Correlation Between Clinical Symptoms and Actual Pulp Inflammation

From a clinical perspective, it would be most helpful to the clinician to be able to diagnose pulp conditions from a profile of symptoms with which a patient presents. If symptoms are not conclusive, a number of objective tests should aid the clinician in reaching a definitive diagnosis of the pulp pathological status. In actuality, such combinations of subjective and objective findings are frequently insufficient in reaching definitive diagnosis of the status of the dental

pulp. This is particularly true in cases of vital inflamed pulp, where it is difficult for the practitioner to determine clinically whether the inflammation is reversible or irreversible.

Many practitioners rely on painful symptoms to determine the status of the pulp. Several studies have examined this question in some detail. Several classic studies were performed in which the subjective and objective clinical findings related to carious teeth were recorded prior to extracting the teeth and examining them histologically. The underlying hypothesis in these studies was that the more severe the clinical symptoms, the more intense pulp inflammation and destruction was evident histologically. These studies showed that in the vital pulp, clinical symptoms generally did not correlate with histopathological findings.[105,189,262] Furthermore, carious pulp exposure was associated with severe inflammatory response or liquefactive necrosis, regardless of symptoms (Fig. 14.9). These histological changes ranged in extent from being present only at the site of the exposure to deep into the root canals.[262] In a few studies, prolonged or spontaneous severe symptoms were associated with chronic partial or total pulpitis, or pulp necrosis.[68,262] However, in these as well as in other studies, it was common to find cases with evidence of severe inflammatory responses including partial necrosis histologically, but with little or no clinical symptoms—the so-called "painless pulpitis."[68,105,189,262] Moreover, the frequency of nerve fibers,[252] and the vascularity[253] in inflamed pulp does not correlate with clinical symptoms in primary and permanent teeth. It has been reported that the incidence of painless pulpitis that leads to pulp necrosis and asymptomatic apical periodontitis is about 40% to 60% of all pulpitis cases.[188] More recently, one study[248] documented the clinical and histopathological findings and showed that the clinical diagnosis of normal pulp/reversible pulpitis match the histologic diagnosis in 96.6% of the time and of irreversible pulpitis 84.4%, respectively. They also noted that the inflammation and bacterial infection was often localized to the coronal pulp when the tooth responded as irreversible pulpitis. This is critical for vital pulp therapy (VPT), as the radicular pulp is maintained in that procedure. However, this study was retrospective, based on one dentist's subjective definitions of reversible and irreversible pulpitis with no standardization of testing. Asymptomatic irreversible pulpitis was not included. Moreover, cases with signs and symptoms of symptomatic irreversible pulpitis and normal periapex respond well to emergency pulpotomy[77] and to VPT with a tricalcium silicate.[166]

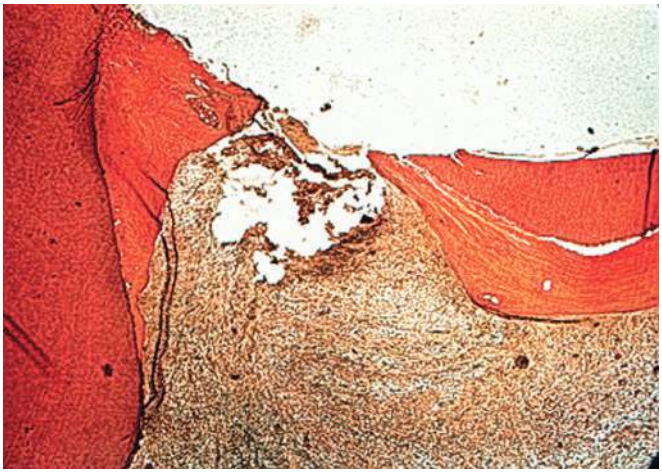


Fig. 14.9 Histological photomicrograph of a molar with carious pulp exposure. The exposure had been capped but had failed and the patient presented with symptoms. The photomicrograph shows an area of necrosis and extensive inflammation throughout the coronal pulp. (Courtesy Dr. Larz Spangberg at the University of Connecticut, with permission.)

Objective clinical findings are essential for determining the vitality of the pulp and whether the inflammation has extended into the periapical tissues. Lack of response to electric pulp testing is generally indicative that the pulp has become necrotic.[247,262] Thermal pulp testing is valuable in reproducing a symptom of thermal sensitivity, and in allowing the practitioner to assess the reaction of the patient to a stimulus and the duration of the response. However, pulp testing cannot determine the degree of pulp inflammation.[68,262] These studies show that irreversible pulp inflammation can be diagnosed with some certainty only in cases where, in addition to being responsive to pulp testing, the pulp develops severe spontaneous symptoms. Pulp necrosis could be predictably diagnosed by a consistent negative response to pulp tests, preferably to both cold and electrical tests to avoid false responses.[234,235] Recent studies show the accuracy of traditional pulp testing to be about 82% to 84%.[179]

The lack of correlation between the histological status of the pulp and clinical symptoms may be explained by recent advances in the science of pulp biology. In the last few decades, studies have shown that numerous molecular mediators may act in synchrony to initiate, promote, and/or modulate the inflammatory response in the dental pulp. The nature and quantity of these inflammatory mediators cannot be determined from histological analysis, without the use of very specialized staining techniques and molecular assays. Many of these molecular mediators tend to reduce the pain threshold, either directly by acting on peripheral nerve cells or through promoting the inflammatory process. Thus a number of these mediators were shown to be elevated in human pulp diagnosed with painful pulpitis. These mediators include prostaglandins,[53,251] the vasoactive amine bradykinin,[163] tumor necrosis factor alpha,[151] neuropeptides such as SP,[32] CGRP and neurokinin A,[14] and catecholamines.[207] In fact, it was even shown that when patients have painful pulpitis, the crevicular fluid related to the affected teeth has significantly increased neuropeptides compared to the levels in contralateral teeth.[14]

It has also been determined that peripheral opioid receptors are present in the dental pulp,[127] and these could play a role in helping us understand why many cases with irreversible pulpitis are asymptomatic. As noted before, carious teeth are frequently not associated with significant symptoms. However, they still have significant amount of inflammation. The pulp in teeth with mild to moderate caries has increased neuropeptide Y,[73] and its Y1 receptor,[74] compared to that in normal teeth. Neuropeptide Y is a neurotransmitter for sympathetic nervous system and is thought to act as a modulator of neurogenic inflammation. Likewise, the levels of VIP, although not its receptor VPAC1, seemed to increase in the pulp of moderately carious teeth.[72]

With recent advances in molecular biology, efficient detection of hundreds of molecular mediators simultaneously

by their gene expression has become a reality. Current research seeks to examine which genes are specifically expressed or upregulated in the pulp, in response to the carious lesion. In this regard, preliminary studies have shown that various cytokines and other inflammatory mediators are upregulated underneath a carious lesion, in a manner that correlates with the depth of caries.[184] Gene microarrays have been used in several studies to obtain an accurate mapping of candidate genes that show elevated expression in inflamed pulp and the odontoblastic cell layer.[185,216,217] In addition, research has revealed the differential expression of microRNAs (miRNAs) in healthy and diseased dental pulp,[326] and the increase in viral microRNA in inflamed pulp, to correlate with levels of inflammatory mediators.[325] miRNAs are noncoding RNA molecules that regulate gene expression in complex inflammatory responses and may eventually assist in clinical predictions of pulp status. Therefore more accurate chair-side diagnostic methods are potentially feasible to develop; particularly, sampling can be made from crevicular fluid, dentinal fluid, or the pulp directly. For this reason, more research is needed to determine the key mediators that would predict survival or degeneration of the dental pulp in difficult diagnostic cases.

Dentin Hypersensitivity and Its Management

Dentin hypersensitivity is a special situation in which a significant, chronic, pulp pain arises, which does not seem to be associated with irreversible pulp pathosis, in most cases. Dentin hypersensitivity is characterized by brief sharp pain arising from exposed dentin in response to stimuli (typically thermal, evaporative, tactile, osmotic or chemical) that cannot be ascribed to any other form of dental defect or pathology.[111] Facial root surfaces in canines, premolars, and molars are particularly affected, especially in areas of periodontal attachment loss. Dentin hypersensitivity may be related to excessive abrasion during tooth brushing, periodontal disease, or erosion from dietary or gastric acids,[2,3,49] and may be increased following scaling and root planning.[49,308] The dentin is hypersensitive most likely due to the lack of protection by cementum, loss of smear layer by acidic dietary fluids, and the hydrodynamic movement of fluid in dentinal tubules.[5,35] The degree of inflammation in the pulp in cases of dentin hypersensitivity is not well characterized because the condition is usually not severe enough to warrant tooth extraction or endodontic therapy. However, patent dentinal tubules are present in areas of hypersensitivity (Fig. 14.10)[320] and may result in increased irritation and localized reversible inflammation of the pulp at the sites involved.

The application of neural modulating agents such as potassium nitrate,[183] or tubule blocking agents such as strontium chloride, oxalates, or dentin bonding agents (Fig. 14.11),[5,224] usually alleviates the condition, at least temporarily. However, the placement of passive molecules or crystals may provide only temporary relief, thus there has been the need to provide biocompatible materials that bond to the root surface, in order to provide a more lasting solution. One such material was a calcium sodium phosphosilicate bioactive glass,[172] which was developed into a commercial product (SootheRx, NovaMin Technology Inc., Alachua, FL). Another product uses a combination of a calcium oxalate and an acid-etched bonding material to seal the dentinal tubules (BisBlock, Bisco Inc., Schaumberg, IL). A concern has been raised that the acidic pH during etching may cause dissolution of the oxalate crystals, thus interfering with the effectiveness of the material,[315] However, one study found that BisBlock and two other products (Seal&Protect, Dentsply DeTrey GmbH, Konstanz, Germany, and Vivasens, Ivoclar Vivadent AG, Schaan, Liechtenstein) were effective compared to placebo, several weeks after treatment,[220] In the long term, the development of smear layer, such as from tooth brushing, dentin sclerosis, reactionary dentin, and the blockage of tubules with large endogenous macromolecules, are all thought to reduce the problem[225] (illustrated by Dentin Hypersensitivity animation at www.PathwaysofthePulp.com). A practice-based randomized clinical trial compared the effectiveness of nondesensitizing toothpaste (Colgate Cavity Protection Regular, Colgate-Palmolive, United States), with desensitizing toothpaste (Colgate Sensitive Fresh Stripe, Colgate-Palmolive, United States) and a professionally applied desensitizing agent (Seal and Protect, Dentsply, United States).[95] The findings showed a significant reduction of dentin hypersensitivity in the desensitizing therapies compared to the nondesensitizing group. In fact, the reduction was much more significant than that obtained by the professionally applied desensitizing agent over a 6-month period. Contemporary desensitizing agents that have been found to be effective include nano-hydroxyapatite,[66] laser treatment, HEMA and glutaraldehyde primer, 3% oxalate, or home products containing 8% arginine/NaF or 5% potassium nitrate.[93]

Pulp Reactions to Local Anesthetics

An intact PBF is critical for maintaining the health of the dental pulp. Because the dental pulp is enclosed in a rigid chamber and is supplied by few arterioles through the apical foramina, it cannot benefit from collateral circulation or volumetric changes that compensate for changes in blood flow in other soft tissues. Furthermore, reduction in blood flow has the compounding effect of reducing the clearance of large molecular weight toxins or waste products,[223] resulting in irreversible pulp pathosis. Vasoconstrictors are added to local anesthetics to enhance the duration of anesthesia. However, vasoconstrictors in local anesthetics could negatively impact the health of the pulp if they reduce blood flow, particularly if the pulp is inflamed preoperatively. Earlier studies have documented that vasoconstrictors in local anesthetics do reduce PBF in experimental animals when administered by infiltration and nerve block (Fig. 14.12),[142] and that this effect was more severe with periodontal ligament injections (Fig. 14.13).[141] More recently, clinical trials were conducted in which subjects were given infiltration of different local anesthetics with or without epinephrine at a concentration of 1:100,000 and the PBF was measured by the laser Doppler flowmetry. In groups that received the epinephrine, there were consistently significant reductions in PBF,[6,51,198] even if the infiltration was palatal to maxillary premolars.[237] Interestingly,

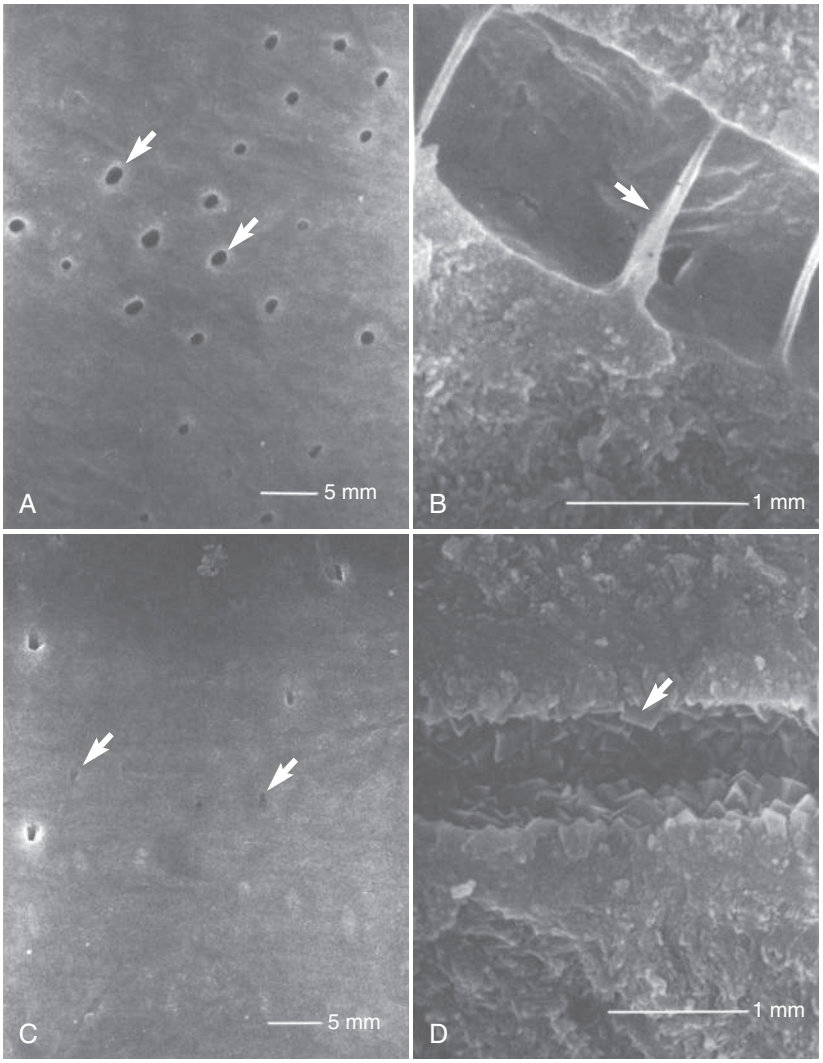


Fig. 14.10 Scanning electron microscope (SEM) image of an exposed dentin surface of a hypersensitive area. **A,** A large proportion of dentinal tubules *(arrows)* are seen to be open. **B,** SEM image of a fractured dentinal tubule of a hypersensitive area. The lumen of the dentinal tubule is partitioned by membranous structures *(arrow)*. **C,** SEM image of exposed dentin surface of a naturally desensitized area. The lumens of dentinal tubules *(arrows)* are mostly occluded, and the surface is extremely smooth. **D**, SEM image of a fractured dentinal tubule of a naturally-desensitized area. Rhombohedral platelike crystals of from 0.1 to 0.3 μm *(arrow)* are present. (Reprinted with permission from Yoshiyama M, Masada J, Uchida A, Ishida H. Scanning electron microscopic characterization of sensitive vs. insensitive human radicular dentin. *J Dent Res* 68:1498–1502, 1989.)



Fig. 14.11 Smear layer treated with 30% dipotassium oxalate for 2 minutes plus 3% monopotassium and monohydrogen oxalate for 2 minutes. Dentin surface is completely covered with calcium oxalate crystals. **(Original magnification ×1900.)** (Reprinted with permission from Pashley DH, Galloway SE. The effects of oxalate treatment on the smear layer of ground surfaces of human dentine. *Arch Oral Biol* 30:731–737, 1985.)

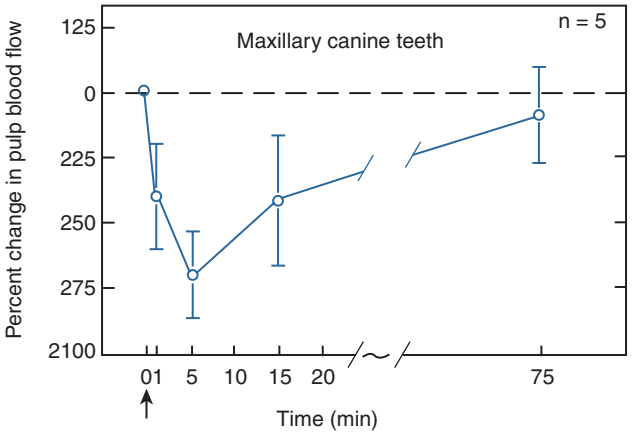


Fig. 14.12 Effects of infiltration anesthesia (i.e., 2% lidocaine with 1:100,000 epinephrine) on pulpal blood flow in the maxillary canine teeth of dogs. There is a drastic decrease in pulpal blood flow soon after the injection. *Arrow* indicates the time of injection. *Bars* depict standard deviation. (Reprinted with permission from Kim S, Edwall L, Trowbridge H, Chien S. Effects of local anesthetics on pulpal blood flow in dogs. *J Dent Res* 63:650–652, 1984.)

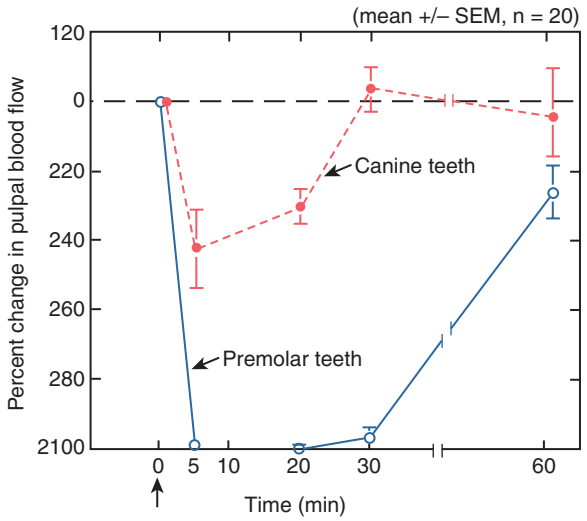


Fig. 14.13 Effects of ligamental injection (i.e., 2% lidocaine with 1:100,000 epinephrine) on pulpal blood flow in the mandibular canine and premolar teeth of dogs. Injection was given in the mesial and distal sulcus of premolar teeth. Injection caused total cessation of pulpal blood flow which lasted about 30 minutes in the premolar teeth. *Arrow* indicates time of injection. (Reprinted with permission from Kim S. Ligamental injection: a physiological explanation of its efficacy. *J Endod* 12:486–491, 1986.)

in one study the reduction in PBF with epinephrine infiltration was more that the reduction in gingival blood flow, and did not return to baseline values after 1 hour of injection.[6] Similar reductions in PBF were reported when inferior alveolar nerve block injections of lidocaine and 1:100,000 or 1:80,000 epinephrine were administered.[209] It is important to note a limitation of studies using laser Doppler flowmetry, which is that a large proportion of the signal measured may be from sources other than the dental pulp.[236,271] Thus the monitoring of minor changes in PBF should be interpreted with caution, particularly if the rubber dam or a similar barrier was not used.[104] Human studies showed that intraosseous anesthesia reduced PBF in molars by 60% after 1 minute, which returned to normal within 45 minutes. A similar reduction was not observed for the blood flow in canines.[309] Taken together, these findings suggest that local anesthesia with vasoconstrictors may compromise the inflamed pulp's ability to recover from inflammation, particularly if it is severely inflamed or if the tooth is subjected to extensive restorative procedures or acute trauma, and if the anesthetic is delivered via a periodontal ligament or an intraosseous route.

Intrapulpal anesthesia is often used as a last resort, when pulp anesthesia is insufficient during root canal therapy. The effect of intrapulpal anesthesia on the pulp in these cases is not considered, as the pulp will be removed. However, occasionally VPT is indicated. One study has shown that intrapulpal anesthesia can be used in these cases, with no clinical differences on follow-up of over 24 weeks between the groups that did or did not receive intrapulpal anesthesia, and when given, in the groups where the anesthetic contained or did not contain epinephrine.[287] However, clearly this injection would have to be performed with caution.

Pulp Reactions to Restorative Procedures

A large body of literature exists on the effects of restorative procedures on the dental pulp. This topic understandably, has been important for practicing dentists for many years. Restorative procedures are performed primarily to treat an infectious disease, dental caries, which itself causes significant irritation of the pulp. They may also be performed to help restore missing teeth, correct developmental anomalies, and address fractures, cracks, failures of previous restorations, or a myriad of other abnormalities. One key requirement of a successful restorative procedure is to cause minimal additional irritation of the pulp so as not to interfere with normal pulp healing. When pulp vitality is to be maintained during a restorative procedure, then a provisional diagnosis of reversible pulpitis rather than irreversible pulpitis must preexist. Therefore it would be most desirable to perform a minimally traumatic restorative procedure, which would not potentially convert the diagnosis to irreversible pulpitis. As was discussed previously, irreversible pulpitis may present clinically with severe spontaneous postoperative pain, but it may also be asymptomatic, leading to the asymptomatic demise of the pulp. The additive effects of restorative procedures are particularly critical in borderline cases, such as those of moderately symptomatic teeth with deep caries but no pulp exposure. There are still many factors whose influence on the response of the dental pulp to cumulative effects of caries, microleakage, restorative procedures, and materials is not well understood. It is generally accepted that the effects of pulp insults, be they from caries, restorative procedures, or trauma, are cumulative. That is, with each succeeding irritation, the pulp has a diminished capacity to remain vital. As a part of informed consent, the clinician is often faced with the task of outlining possible risks of restorative treatment. One study addressed the fate of pulps beneath single unit metal-ceramic (MC) crowns or MC bridge abutments.[50] Patients who had received either treatment were invited to attend a recall appointment that involved both clinical and radiographic examinations. One hundred and twenty-two teeth with preoperatively vital pulps treated with single unit MC crowns and 77 treated as bridge abutments were examined. The mean observation period was 14 years for the former and 15.6 years for the latter. Pulp necrosis had occurred in 15.6% of the teeth treated with single unit crowns, while 32.5% of the pulps in the bridge retainer groups had become necrotic. There was a significantly higher percentage of pulp necrosis in anterior teeth that served as bridge abutments (54.5% of anterior abutment teeth examined). In general, however, the available evidence indicates that the effects of dental procedures on the pulp depend on the following factors.

THE DEGREE OF INFLAMMATION OF THE PULP PREOPERATIVELY

As stated previously, the dental pulp is compromised in its ability to respond to external irritants because it is enclosed in a noncompliant environment, and because it lacks collateral circulation. Thus the more severe the pulp is

inflamed, the less it will be able to respond to further irritation, such as in the form of restorative procedures.[160]

Most research studies designed to evaluate the effects of restorative procedures (or materials) on the pulp are conducted on human or experimental animal teeth with normal pulp tissue. Furthermore, many of the animal research projects have been performed on anesthetized animals without the use of local anesthesia, which as stated previously, reduces PBF. Therefore the results of these studies may not reveal the true clinical effects of these procedures when the pulp is already inflamed by the carious lesion and PBF is reduced by local anesthetic with vasoconstrictors. In a study that evaluated the response of the pulp to capping procedures as a function of duration of exposure, it was shown that the pulp responds favorably to exposures of up to 24 hours after exposure, but not as favorably after longer periods of exposure to oral environment.[60] It may be that the longer exposure periods lead to the formation of a bacterial biofilm, which is difficult for the pulp immune responses to eliminate, and/or the extension of the infection so deep into the pulp to preclude healing. This is relevant in cases of aseptic mechanical exposures or teeth where the pulp is exposed by traumatic injuries for a brief duration. In these cases, the pulp usually responds favorably to VPT procedures.[7] Models of standardized pulp inflammation with chronic caries are not commonly used in determining the effects of dental procedures. Older clinical studies show an unfavorable long-term outcome of capping cases with carious pulp exposures[18,115]; however, newer studies in which tricalcium silicates were used show more favorable results in these cases.[28,166,186,230]

In the absence of severe spontaneous symptoms or pulp exposure, as indicated previously, the clinician currently cannot determine accurately the degree of preoperative pulp inflammation. Thus every effort should be done to minimize added irritation during restorative procedures, as it is possible that excessive irritation could convert the inflammatory status of the pulp from a reversible to an irreversible condition. In addition, the patient should always be advised of the possibility of pulp degeneration and the importance of follow-up.

THE AMOUNT OF PHYSICAL IRRITATION CAUSED BY THE PROCEDURE

Physical irritation during restorative procedures such as from heat, desiccation, or vibration may adversely affect the dental pulp.

Heat

Restorative procedures such as cavity or crown preparation or curing of resins during direct fabrication of provisional restorations[288] may cause significant increases in pulp temperatures. It has been shown using primate models that an intrapulpal temperature rise of 10°C causes irreversible pulp pathosis in 15%, and a 20°C rise caused pulp abscess formation in 60% of teeth evaluated.[322] Several other older studies documented burns or severe inflammation in the pulp when cavity or crown preparations were performed without coolants (Figs. 14.14–14.16). However, a more recent study in which gradual controlled heat application over a large area of the intact occlusal surface of human

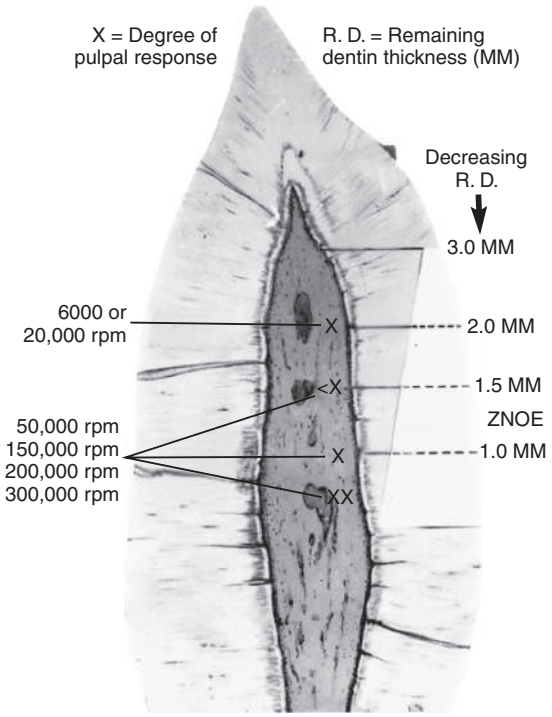


Fig. 14.14 With adequate water and air spray coolant, the same cutting tools, and a comparable remaining dentin thickness, the intensity of the pulpal response with high-speed techniques (i.e., decreasing force) is considerably less traumatic than with lower speed techniques (i.e., increasing force). (Reprinted with permission from Stanley HR, Swerdlow H. An approach to biologic variation in human pulpal studies. *J Prosthet Dent* 14:365, 1964.)

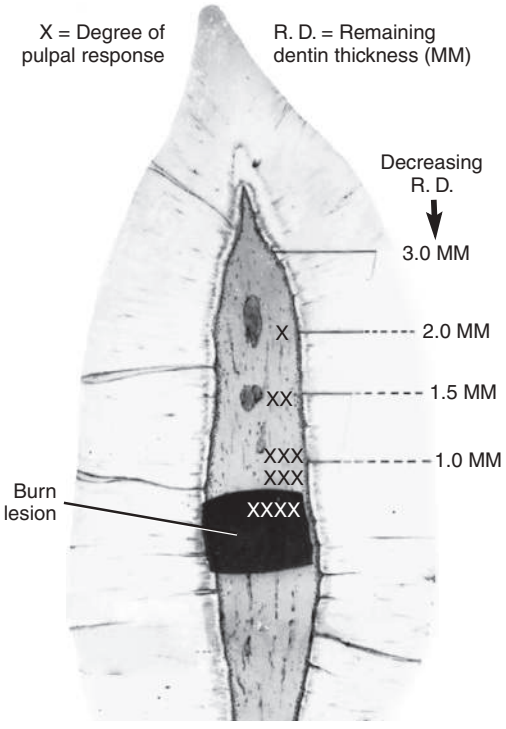


Fig. 14.15 Without adequate water coolant, larger cutting tools (e.g., no. 37 diamond point), create typical burn lesions within the pulp when the remaining dentin thickness becomes less than 1.5 mm. (Reprinted with permission from Stanley HR, Swerdlow H. An approach to biologic variation in human pulpal studies. *J Prosthet Dent* 14:365, 1964.)

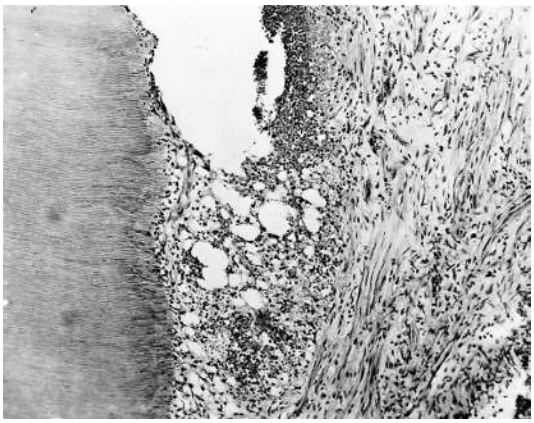


Fig. 14.16 Burn lesion with necrosis and expanding abscess formation in a 10-day specimen. Cavity preparation dry at 20,000 rpm, with remaining dentin thickness is 0.23 mm. (Reprinted with permission from Swerdlow H, Stanley HR, Jr. Reaction of the human dental pulp to cavity preparation. I. Effect of water spray at 20,000 rpm. *J Am Dent Assoc* 56:317–329, 1958.)

unanesthetized teeth was employed, failed to corroborate these earlier findings.[15] In this study, an increase of intrapulpal temperature of about 11°C followed by 2- to 3-month evaluation did not show any clinical or histological changes in the pulp of any of the teeth evaluated. A heat increase in rat pulp tissue to 42°C and in vitro raised heat shock protein-70, which are known to be tissue protective, caused changes in alkaline phosphatase and gap junction proteins, that were reversed to normalcy a few hours later.[9] By contrast, heat applied in deep cavity preparations, prepared atraumatically in human teeth, in another study, caused histologic changes that were dependent on the proximity of the heat source to the pulp.[208] It was common in that study to see a loss of odontoblasts or their aspiration into the dentinal tubules. In cases where the cavity floor was less than 0.5 mm from the pulp, areas of coagulation necrosis could be seen, although the patients remained asymptomatic for the 1-month duration of the study. The measurement of heat in the tooth being prepared in areas other than the site of tooth preparation, occasionally shows reduction in temperature,[96] presumably because of the poor conductive properties of dentin and the cooling effect of compressed air of the high-speed handpiece. Furthermore, cavity and crown preparations include several other irritating stimuli, such as desiccation, severance of odontoblastic processes, vibration, and smearing of bacterial irritants onto the surface of dentin. Therefore taken together, these findings suggest that the transient increase in temperature to levels relevant to modern dental procedures on its own may not be the culprit in inducing pulp changes. Rather, the synergistic application of excessive heat with other irritation factors and its proximity to the pulp may induce pathological changes.

Desiccation

Desiccation during cavity and crown preparation has long been known to cause aspiration of odontoblastic nuclei into dentinal tubules and pulp inflammation.[33] One study showed that as little as 30 seconds of continuous air drying of class V cavities in human molars with uninflamed pulp caused significant displacement of odontoblastic nuclei, pulp inflammation and even areas of necrosis related to the areas that were dried.[59] However, another study showed that the effects of desiccation are transient and that within 7 to 30 days, there is autolysis of the aspirated cells and formation of reactionary dentin.[34] The pulp, in cases with aspirated odontoblasts following desiccation for 1 minute, was not sensitive to clinical scraping with an explorer. The sensitivity was restored with rehydration of the cavities and was increased in other cases where pulp inflammation was induced by microbial contamination.[168] In this study, despite the lack of sensitivity in desiccated cavities, neural elements were seen histologically to be pushed into the tubules, like the odontoblastic nuclei. The disruption of the odontoblastic layer and peripheral neural elements in the pulp with desiccation was also observed in a rat model using axonal transport of radioactive protein.[39] One clinical observational study showed that air cooling without water during crown and bridge preparations caused minimal damage to the pulp clinically or radiographically.[174]

Biological and Chemical Irritation

Dental caries is clearly a disease in which microorganisms and their virulence factors constantly irritate the pulp, even at the early stages, long before pulp exposure.[36] However, despite the elimination of visible caries during cavity preparation, the cavity floor is undoubtedly left with some contamination by caries bacteria. While the rubber dam should be used with any cavity preparation to prevent cavity contamination with salivary microorganisms, the use of water coolants allows the cavity to be contaminated with bacteria from water lines. Concerns about residual cavity contamination prompted some to use cavity disinfection with some caustic chemicals. Chemicals such as hydrogen peroxide, sodium hypochlorite, or calcium hydroxide solutions have been proposed for this purpose, although they may exert a toxic effect.[56] An earlier study showed that the amounts of residual bacteria following adequate restoration are not significant.[190] Once dentin is exposed, there is a constant outward flow of dentinal fluid that minimizes the inward flow of any noxious agents.[310] This may aid in the reduction of irritation from residual microbial factors in dentinal tubules.

In contemporary practice, most chemical irritation during restorative procedures results from the application of etching agents, especially strong acids, in the form of total dentin etch, particularly if capping of exposed pulp is performed.[98,219] Etching is performed to remove the smear layer, to promote physical adhesion of bonding agents to dentin by forming resin tags in the dentinal tubules, and to permeate the newer unfilled resin primers into the unmineralized surface layer of collagen to form the so-called "hybrid layer."

If the cavity is relatively superficial and is adequately sealed with a restorative resin, then the etching of dentin is probably not detrimental to the pulp because of the narrow diameter of dentinal tubules and their low density in peripheral dentin.[30] In fact, one study documented that histological evidence of bacteria in human cavities restored with composite was significantly less if the cavity had been etched with phosphoric acid than if it was etched with 17% EDTA or nonetched.[197] Pulp inflammation in this study was not correlated with the etching treatments, but with

bacterial presence; thus in cases of etching with phosphoric acid, if bacteria were also present, severe pulp inflammation and necrosis could be seen.

Self-etching formulations have become popular because they eliminate the separate etching step involved in total-etch procedures. Some have speculated that the bonding of self-etching systems may be poorer than total-etch systems because of the weaker acidity of the acidic primers of self-etching systems compared to total-etch systems.[30] However, studies have shown no significant differences between the two adhesive systems in long-term in vitro bond strength, postoperative sensitivity,[232] long-term in vivo degradation,[153] or long-term in vitro bond strength.[11] One clinical study showed no differences between the two systems with respect to bacterial leakage and the inflammatory response in the pulp.[202] The most important variable that affected the pulp in this study was the amount of bacterial leakage with either system.

Other factors that may contribute to pulp irritation during resin placement from chemical/biologic irritants include unpolymerized monomer and polymerization shrinkage. Higher concentrations of monomeric resin components were shown to exert an inhibitory effect on T-lymphocytes and spleen cells,[132] as well as monocytes/macrophages[162,240,241,261] in vitro. These components may leach directly into the pulp in deep cavities and cause chemical irritation.[118] Shrinkage during polymerization of composites may induce internal stresses on dentin and create voids that allow microleakage. Shrinkage of resins is estimated to range from 0.6% to 1.4% and should be minimized during placement by incremental curing and possibly starting the restoration with flowable resins.[30]

Recent studies have shown that salivary or dentin matrix-based metalloproteinases or salivary esterases may be released during etching and may degrade the collagen in the hybrid layer, thereby degrading the dentin/composite bone.[116,182] In vitro and in vivo research showed that this detrimental effect could be reduced and the stability of the hybrid layer maintained with the use of chlorhexidine, galardin, benzalkonium chloride, or quaternary ammonium methacrylates.[182]

In summary, the available evidence indicates that chemicals involved in modern restorative procedures may irritate the pulp if placed directly on an exposure, if there is microbial leakage along the tooth/restoration interface, or if there is degradation of the hybrid layer that seals the restoration dentin interface.

THE PROXIMITY OF THE RESTORATIVE PROCEDURES TO THE DENTAL PULP AND THE SURFACE AREA OF DENTIN EXPOSED

It has been known for several decades that, as the carious lesion progresses towards the pulp, particularly when the RDT is less than 0.5 mm, there is an increasingly severe pulp reaction, with a greater likelihood of the pulp undergoing irreversible pathosis.[245] The diameter and density of dentinal tubules increase closer to the pulp (Fig. 14.17). Based on the dentinal tubule density at the DEJ (about 65,000/mm^2) and the pulp (about 15,000/mm^2),[90,94] it was estimated that the area occupied by tubule lumina at the DEJ was 1% of the total surface area at the DEJ and 22% at the pulp.[225] Thus it is not surprising that several studies have shown that pulp inflammation in response to restorative procedures increases with the reduction in RDT.[1,61,196] One study examined the differential effects on the rat pulp of the preparation method, RDT, coolant, drill speed, conditioning with EDTA, and filling materials.[195] Subsequent to the cavity preparations a tooth slice was obtained and maintained ex vivo as an organ culture for up to 2 weeks. The results showed that the RDT was the most important factor on pulp injury.

With the passage of time following cavity preparation, there is a reduction in the permeability of RDT.[227] This may be due to the rapid deposition of reactionary dentin, the migration of large proteins into the tubules, and/or the diminution of tubule diameter as dentin becomes more sclerotic. Using a primate model, it was shown that the basic rate of secondary dentin deposition was about 0.8 μm/day and that this rate increased to an average of 2.9 μm/day following restorative procedures. Interestingly, in this study, dentin deposition was also more rapid next to shallow cavities than deep cavities[312]; however, another

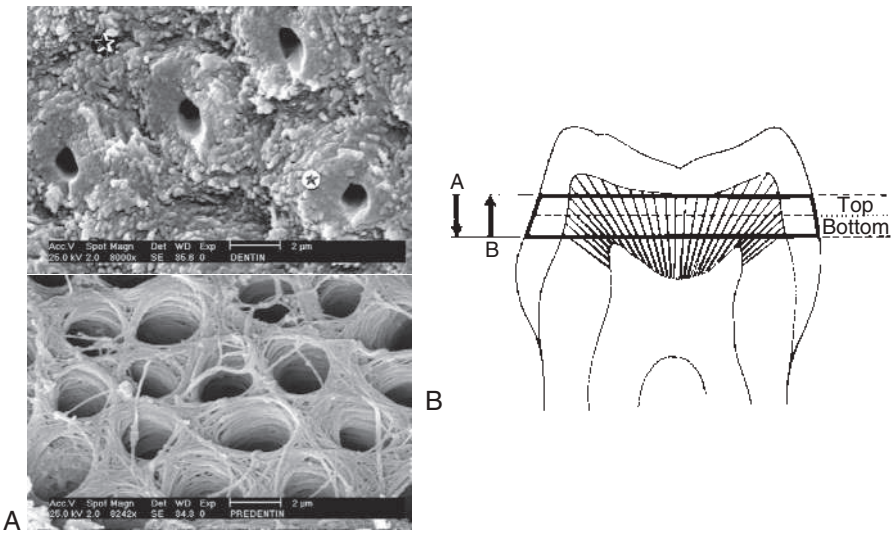


Fig. 14.17 Schematic of convergence of tubules toward the pulp. **A,** Periphery of the dentin. Most surface area is occupied by intertubular dentin (☆), with a few tubules surrounded by hypermineralized peritubular dentin (✪). **B,** Near the pulp, the increase in tubule diameter has occurred largely at the expense of the peritubular dentin. This substrate has high protein content. As the remaining dentin is made thinner (from A to B), the permeability of the dentin increases, because both diameter and density of dentinal tubules are increased. (Reprinted with permission from Bouillaguet S. Biological risks of resin-based materials to the dentin-pulp complex. *Crit Rev Oral Biol Med* 15:47–60, 2004.)

study showed that total reactionary dentin deposited was thicker in deeper and wider cavities.[194]

Clinically, it has been observed that postoperative sensitivity is common with many restorative procedures. Following resin composite restorations on patients, it was shown that postoperative sensitivity was related to the depth of the cavity but not to the presence or absence of liners or bases.[302]

In addition to the depth and/or the width of a large cavity preparation, a crown preparation exposes more dentinal tubules to microbial or chemical irritation. During crown fabrication, there are added irritation factors such as the length of time of preparation, impression techniques, and the imperfect adaptation of temporary restorations that cause microleakage during the temporization period. Due to the precise engineering requirements of some restorations, some providers may be inclined to reduce the coolant during crown preparations steps, such as finalizing the finishing lines. However, crown preparations without coolants have been shown to dramatically reduce PBF in an animal model (Fig. 14.18).[143] Air coolant appears to be adequate, at least in part.[174] In general, long-term outcome studies have documented that the incidence of pulp necrosis following crown placement ranges from 10% to 50%.[50,85,303]

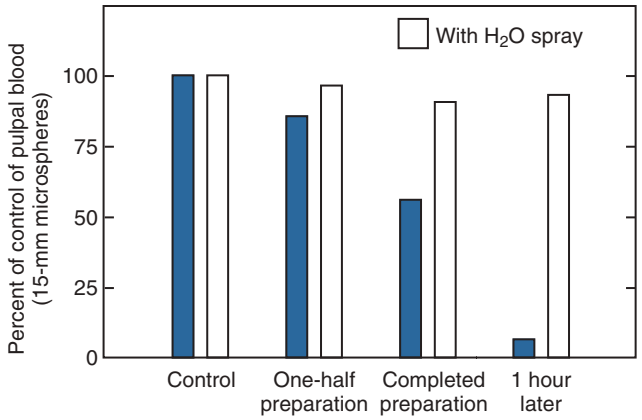


Fig. 14.18 Effects of crown preparation in dogs, with and without water and air spray (at 350,000 rpm) on pulpal blood flow. Tooth preparation without water and air spray caused substantial decrease in pulpal blood flow, whereas that with water and air spray caused insignificant changes in the flow (Reprinted with permission from Kim S, Trowbridge H, Suda H. Pulpal reaction to caries and dental procedures. In: Cohen S, Burns RC, editors. *Pathways of the pulp,* ed 8, St. Louis, 2002, Mosby, pp 573–600.)

THE PERMEABILITY OF DENTIN AND THE ODONTOBLASTIC LAYER BETWEEN THE AREA BEING RESTORED AND THE PULP

The permeability of dentin plays an important role in the ingress of potential irritants to the pulp. It is clear from research done in the last three decades that dentin is not uniformly permeable and that permeability depends on factors such as the location within the same tooth, the age of the patient, and the presence of pathological conditions such as dental caries. Fundamentally, the permeability of dentin depends on the collective sum of permeability of individual tubules at a specific site in the tooth. The tubular diameter increases from about 0.6 to 0.8 μm close to the DEJ to about 3 μm at the pulp.[222] Given that bacterial cells are about 0.5 to 1 μm in diameter, it is evident that in deep cavity preparations, particularly when total-etch procedures are employed, bacteria can migrate through the remaining dentin into the pulp.

With age, the width of peritubular dentin increases, causing reduction in tubular lumen or sclerosis. Caries causes demineralization in superficial dentin, which is associated with remineralization and the formation of caries crystals within the tubules of inner undemineralized dentin (Fig. 14.19). This causes a decrease in permeability in dentin subjacent to the carious lesion[224] and could be considered a protective mechanism, as it may delay the progress of the carious lesion.

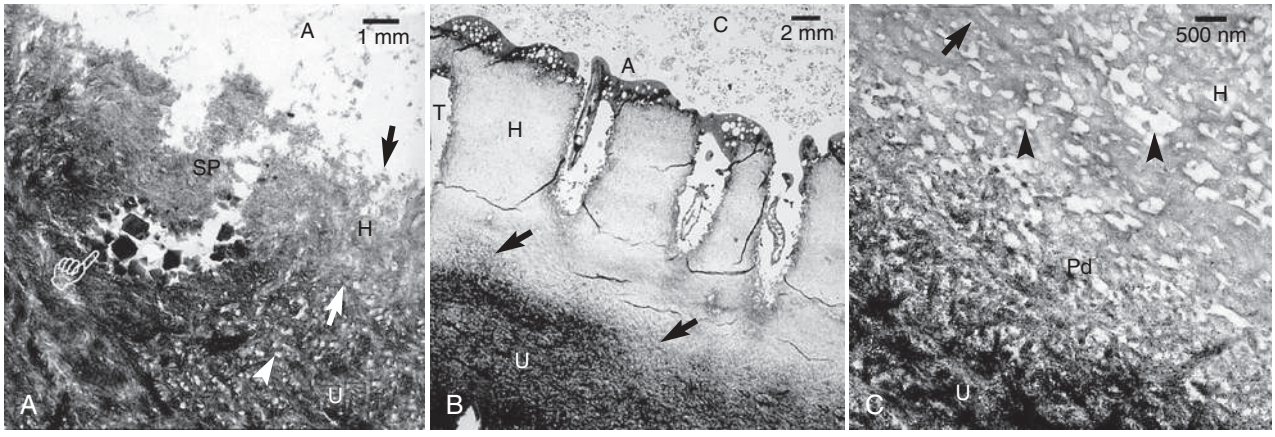


Fig. 14.19 Transmission electron microscope (TEM) images of undemineralized specimens of resin-bonded caries-affected dentin. **A,** Stained TEM image of undemineralized specimens following the application of the self-etch ABF system to caries-affected dentin. The hybrid layer (*H,* between *arrows*) was about 3 μm thick, and the underlying un-demineralized dentin *U* was highly porous *(arrowhead)*. The dentinal tubule was covered with a smear plug *(SP)* and was partially obliterated with large caries crystals (pointer). *A,* Filled adhesive. **B,** Stained section of the total etch Single Bond adhesive bonded to caries-affected dentin. A hybrid layer *H* between 15 and 19 μm thick could be seen, with a partially demineralized zone *(open arrows)* above the undemineralized caries affected dentin *U. C,* Composite; *T,* dentinal tubule. **C,** Higher magnification of the basal part of the unusually thick hybrid layer *H* shown in *A*. Banded collagen fibrils *(open arrow)* were separated by unusually wide and porous interfibrillar spaces *(open arrowheads)*. A partially demineralized zone *(Pd)* was present along the demineralization front. This zone was not seen in phosphoric-acid-etched sound dentin (B). *U,* Undemineralized caries-affected dentin. (Reprinted with permission from Yoshiyama M, Tay FR, Doi J, et al. Bonding of self-etch and total-etch adhesives to carious dentin. *J Dent Res* 81:556–560, 2002.)

It was shown that irritation from cavity preparation increased the permeability of the odontoblastic cell layer only at the site of the cavity preparation.[125,297] In addition to the physical barrier to permeability and the production of reactionary or reparative dentin, the odontoblastic layer may, in fact, contribute to the host response of the dental pulp by expressing important inflammatory mediators[165,218] or recognize bacteria through TLRs.[29,71,80,131]

THE AGE OF THE PATIENT

Resting PBF, as well as the changes in PBF in response to cold application, decreases with age.[121] Age may also be associated with a reduction in pulp neuropeptides.[91] However, studies show no differences between young and old pulp in the regenerative capacity of odontoblast-like cells, nor in the presence of cells positive for class II major histocompatibility complex, heat shock protein 25 or nestin, when subjected to cavity preparation.[137] An examination of normal, young versus old human pulp showed that young pulp had an increased expression of biological factors related to cell differentiation, proliferation, and the immune response; however, older pulp had increased factors related to apoptosis.[293] Nevertheless, PBF in anterior teeth measured with laser Doppler flowmetry showed that it correlates with systolic blood pressure, but not with age, sex, or tooth type.[140] Taken together, these analyses cannot translate to the pulp's ability to deal with irritation and its sequelae. Thus the net result of the ability of the pulp to cope with external stimulation or irritation in humans with advancing age is not clear.

Pulp Reactions to Restorative Materials

The effects of restorative materials on the dental pulp have been investigated and seem to relate directly to the permeability of the associated dentin. The degree of dentin permeability, however, is often variable and is governed by several factors, including age and caries status.[284] The most important variable in dentin permeability to restorative materials is the thickness of the dentin between the floor of the cavity preparation and the pulp.[1]

Given the importance of dentin permeability, there are direct effects of any given restorative material on the pulp that is governed by the composition of the material and associated eluted or degraded products. Unbound components of resin materials and preparative agents such as acid etchants can affect the subjacent pulp by inducing an inflammatory response.[93,98,239] The indirect effects of desiccation and/or demineralization of dentin as well as the direct effects of the material itself when in contact with pulp tissue mediate this inflammatory response. Studies have shown that the certain cytotoxic components of resin monomers (e.g., triethylene glycol dimethacrylate and 2-hydroxyethyl methacrylate) readily penetrate dentin.[87] Similarly, eugenol and components of Ledermix (triamcinolone and demeclocycline) have been shown to pass through dentin into the subjacent pulp.[117,120] In vivo data show that these chemicals have an effect on the pulp; however, the effect seems to be short lived and, in the absence of bacteria, reversible.[23]

The mechanisms whereby restorative materials exert an injurious effect on the dental pulp are varied. Evidence exists that supports direct and, in some instances, prolonged cytotoxicity, stimulation of hypersensitivity reactions, or impairment of the host immune response to bacteria.[258] Some of the components of resin restorations are released at cytotoxic levels after polymerization is completed, leading to chronic stimulation and a resultant prolonged inflammatory response.[86] Furthermore, even subtoxic concentrations of certain agents are capable of eliciting allergic reactions in humans.[119] Primates hyperimmunized with BSA showed significant pulp damage with repeated antigenic challenge in class V cavity preparations, suggesting a role for antigen-antibody complex–mediated hypersensitivity in tissue destruction.[26] In a separate study, exposure to dentin primers elicited a delayed-type hypersensitivity reaction in guinea pigs.[136] These studies, taken together, present a compelling argument for immune-mediated pulp tissue damage subsequent to exposure to restorative materials. Foreign body reactions have also been described in pulps containing extruded globules of resin material.[123,124] Histological examination of such pulps shows macrophages and giant cells surrounding the resin particles. Lastly, resin monomers have been shown to decrease the activity of immunocompetent cells in a dose-dependent manner in in vitro functional assays.[132] While all these effects are documented, their extent and therefore morbidity on the dental pulp is speculative and doubtless does not act solely to effect pulp demise. As previously noted, most restorative materials are placed adjacent to pulps that are previously compromised by bacterial insult and that disease, debridement, and restoration of the tooth have cumulative effects on the dental pulp.

Although pulp irritation is largely considered to be a negative sequela, the irritant potential of certain restorative materials is central to their usefulness in restorative dentistry. Calcium hydroxide is one of the oldest and most widely used medicaments for stimulation of dentinal bridge formation subsequent to microscopic or gross pulp exposure. The low-grade pulp irritation that it induces is important for dentinal bridge formation in exposures.[64,260] The degree of inflammation is dependent on the preparation of calcium hydroxide used. Aqueous suspensions of calcium hydroxide applied to exposed pulps cause superficial necrosis of pulp tissue followed by low-grade inflammatory changes. Within 30 days, the tissue subjacent to the necrotic zone has reorganized and resumed normal architecture. Hard setting calcium hydroxide preparations as well as mineral trioxide aggregates (MTAs) are effective in eliciting dentinal bridge formation with a much smaller to nonexistent necrotic zone.[108] This is preferable in vital pulp therapies such as the Cvek pulpotomy where maintenance of the maximum amount of vital pulp tissue is desirable and the extent of pulp inflammation is minimal (Fig. 14.20).[63] The irritation potential of calcium hydroxide across intact dentin is dependent on factors such as the RDT and permeability. Application of calcium hydroxide to intact dentin appears to induce sclerosis by promoting crystal precipitation within the tubules accompanied by reductions in permeability.[192] However, more recent findings show that calcium hydroxide over intact dentin does not exert a significant clinical advantage and may be lost due to the action of leakage or dentinal fluid.[62]

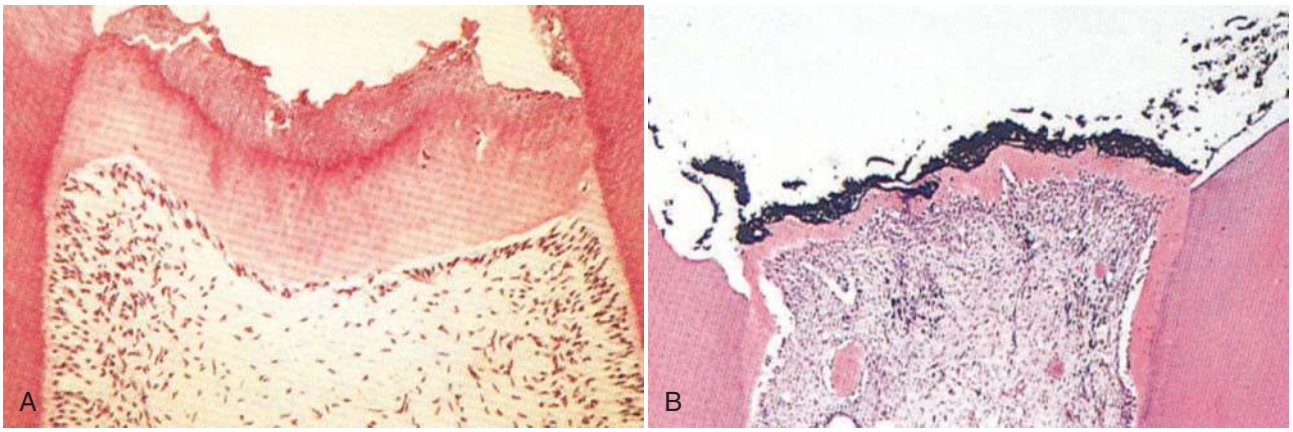


Fig. 14.20 Calcium hydroxide produces an inflammatory response that stimulates dentinal bridge formation. The dentin bridge forms lower in the tooth with calcium hydroxide paste **(A)** versus hard setting calcium hydroxide **(B)**.

While the irritation potential of calcium hydroxide plays a role in its effectiveness, the high pH of this material can act to liberate bioactive molecules from dentin. Numerous authors have demonstrated that dentin matrix proteins like TGF-β1 and adrenomedullin are liberated by both calcium hydroxide and MTA.[97,291] Once liberated, they are able to facilitate hard tissue formation yet again. This offers another explanation for the ability of these materials to induce hard tissue formation in vivo. Acid etching or the use of chelating agents such as EDTA represent a common step in the placement of bonded restorations. The etching process removes minerals and exposes collagen fibrils. As with a high pH, sequestered bioactive molecules are released and can act on pulp cells. These include growth factors and metalloproteinases, which can be inductive for healing but can also degrade the bond of the restoration.[228] EDTA has been shown to be even more effective at extracting noncollagenous proteins from dentin than either calcium hydroxide or MTA, but less so than citric acid.[97,291]

Investigations into the pulp response to glass ionomer materials have shown that both the lowered pH of the material and the release of high concentrations of fluoride can inflict damage to pulp tissues.[135] In direct contact with pulp cells, glass ionomers are toxic.[155] Direct pulp capping studies have shown it to be inferior to calcium hydroxide.[67,112] When applied to intact dentin and in the absence of bacteria, there is a transient inflammatory response and reparative dentin is formed.[57]

In addition to the direct chemical effects of restorative materials, there are indirect factors that contribute to pulp irritation. The technique sensitivity of certain materials predisposes them to faulty bonds to tooth structure that can translate to dentin hypersensitivity, recurrent disease, and pulp inflammation or necrosis. Much attention has been given to the interface created between resin-bonded materials and dentin. During the etching process, the more highly mineralized peritubular dentin is preferentially dissolved, leaving free collagen fibrils and opening lateral tubular branches.[98,99,191] Applied resin infiltrates the exposed collagen mesh creating a layer 5 to 10 μm thick referred to as the hybrid layer.[199] This layer, along with the resin permeating exposed tubules, forms the bond between the resin and dentin. If the preparation is too dry, the collagen fibrils collapse and the resin cannot effectively permeate the mesh, which results in a defective bond. As the optimal degree of hydration of the preparation surface can vary from material to material, resin restoration placement is technique-sensitive. This same principle is applicable to the practice of bonding fractured tooth fragments where the segment has become dehydrated while outside of the mouth. Current protocols recommend rehydration of the segment prior to bonding, thus increasing the mechanical and, presumably, the microbial seal.[84] This is particularly important with a complicated crown fracture where the pulp protection by intact dentin is absent.

Some restorative materials rely on their medicinal properties as well as their ability to seal a cavity preparation. Materials containing zinc oxide and eugenol (ZOE) fall into this category. ZOE is used for a variety of purposes in dentistry largely because of its anesthetic and antiseptic properties. It has been shown to block the transmission of action potentials in nerve fibers and to suppress nerve excitability in the pulp when applied to deep excavations.[295] In addition, ZOE has good adaptation to dentin and inhibits bacterial growth on cavity walls. These properties have made it a favored material for temporary fillings, but not long-term restorations, as ZOE temporaries have been shown to leak after only a few weeks in situ.[323]

Direct Pulp Capping With Bioceramics

The direct capping of pulp exposures is indicated in pulps that were previously healthy and exposed by trauma or dental restorative procedures and, more recently, for pulp exposure where the diagnosis is reversible pulpitis.[4,279,283] This is particularly true in the cariously exposed immature permanent tooth, where maintenance of pulp vitality is crucial to further tooth development. Although calcium hydroxide has historically been the preferred dressing agent on mechanically exposed pulps, the use of MTA, Biodentine, and other bioceramic formulations has recently been proposed, even on carious pulp exposures.[17,28,109,186] Prospective animal studies and human case reports have evaluated the ability of MTA to allow for the formation of a reparative

dentin bridge and to maintain continued pulp vitality.[79,89,148] Although the results are generally favorable, one concern is of tooth discoloration in cases where the gray MTA formulation is used on anterior teeth. Recent studies have indicated that newer bioceramic materials show less potential for discoloration when used as pulp capping materials.[149]

The bioceramic that has been most intensely studied for pulp capping is MTA. In one clinical study, MTA was used as a pulp capping material for carious pulp exposures.[28] Forty patients, ages 7 to 45, who were diagnosed with reversible pulpitis had caries removed using a caries detection dye and sodium hypochlorite for hemostasis. The treatment was performed in two visits to allow the MTA to set up and to confirm pulp sensibility to pulp tests in the second visit. Success was determined radiographically, with subjective symptoms, pulp testing with cold, and continued root formation on immature teeth. Outcomes were measured over a period of up to 9 years postoperatively and showed an overall success rate of 97%, with all the teeth in the immature root group showing success. In another clinical study in which dentists and students participated in the treatment, pulp capping of carious pulp exposures was compared between MTA and calcium hydroxide.[186] At follow-up, 122 cases of patients of mean age of 40 years were available. The success with MTA was 78% and, with calcium hydroxide, success was 60%, a difference that was statistically significant. Within the parameters of these studies, it seems that white MTA is a suitable capping agent for pulp exposures of healthy or reversibly inflamed pulps, including cases with asymptomatic carious exposures. Randomized trials and consensus reports have shown that tricalcium silicates are significantly better than calcium hydroxide as pulp capping agents in carious pulp exposure.[69,157] However, given that the prognosis of root canal therapy in this last category has been shown to be predictably high in many studies, the patient needs to be educated carefully of the options, and consent must be documented.

As previously mentioned, other calcium silicate-based materials are gaining popularity as pulp capping agents and appear to have similar properties and effects as MTA.[12,205] Biodentine deserves special reference because, in addition to the biocompatibility, it has physical properties that are very similar to dentin.[41,154] Studies show no differences between MTA, Biodentine, and other tricalcium silicates like NeoMTA Plus as pulp capping or pulpotomy agents.[134,206,290]

The Use of Hemostatic Agents and Disinfectants on Direct Pulp Exposures

There is still debate as to whether the outcomes of pulp capping for exposure depends on the toxicity of medicaments and materials placed on vital pulp tissues, the ability of these materials to induce mineralization, or their ability to seal the cavity from further bacterial ingress. It is likely a combination of these factors, as was shown in the recent clinical trial with MTA mentioned before.[28] Another factor in the prognosis of direct pulp caps is the ability to control hemorrhage at the exposure site.[273] Given the difficulty in creating a bacteria-free operating environment during tooth preparation, the ideal hemostatic agent also would have the ability to kill bacteria.

One study compared the effects of two hemostatic/disinfectant agents on the healing of experimental pulp exposures created in human third molar teeth and capped with calcium hydroxide.[264] Pulp exposures were made in 45 maxillary wisdom teeth scheduled for extraction for orthodontic reasons. Teeth were randomly assigned to receive hard-setting calcium hydroxide pulp caps after a 30-second surface treatment with one of three agents: 0.9% saline, 2% chlorhexidine, or 5.25% sodium hypochlorite. Although the 7-day saline specimens showed slightly less inflammatory response, there were no statistically significant differences between the groups with respect to all dependent measures over the course of the study. Complete healing was seen in 88% of all specimens at 90 days.

The pulps in these teeth were previously uninjured, and the exposures were made in a clean environment. Carious pulp exposure frequently reveals an inflamed pulp in which it is difficult to stop the bleeding. The original study by Bogen et al. proposed applying a cotton pellet with 5.25% to 6% sodium hypochlorite for 1 to 10 minutes for hemostasis.[28]

Pulp Reactions to Laser Procedures

Numerous studies have been published on the effect of using lasers on enamel, dentin, and pulp. Laser use on hard tissue has been a popular area of research because of the potential benefits of efficiency, reduced sensitivity, disinfection, and precision. There are several different types of laser technologies available that depend on the wavelength, active medium, emission mode, delivery system, power output, and duration of application. The main types available in dentistry today are shown in Fig. 14.21. The CO_2 laser is historically the oldest type used on soft tissues and thus has been the most studied. It has the longest wavelength (10,600 nm). It cannot be delivered in an optic fiber and

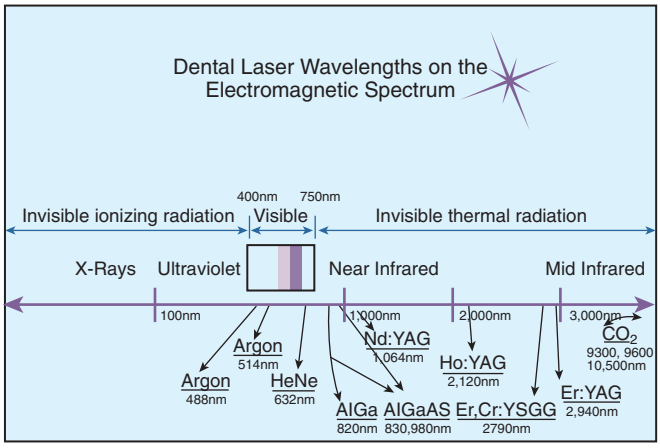


Fig. 14.21 Currently available dental wavelengths on the electromagnetic spectrum. Note all of the wavelengths are nonionizing. (Reprinted with permission from Coluzzi DJ. An overview of laser wavelengths used in dentistry. *Dent Clin North Am* 44:753–765, 2000.)

thus must be used in a hollow tubelike wave guide in continuous gated-pulse mode. This means that the operator does not feel a solid resistance when using this laser. Er:YAG, Nd:YAG, or Ho:YAG lasers all have an active medium of a solid crystal of yttrium-aluminum-garnet, which is impregnated in erbium, neodymium, or holmium, respectively. Er:YAG has a wavelength of 2940 nm and is delivered using a solid optic fiber. It has a high affinity for water and hydroxyapatite, which means it can be used for removal of caries and cutting dentin with coolant. It can also be used on soft tissue. The Ho:YAG laser has a wavelength of 2120 nm, has a high affinity for water but not to tooth structure, and thus is used primarily for soft-tissue surgery. Nd:YAG laser is also delivered fiberoptically, has a wavelength of 1064 nm, and has been used extensively in dentistry because it has a high affinity for water and pigmented tissues and offers good hemostasis; therefore it is used extensively in surgery.[54] In addition, there are some low power output lasers such as HeNe or helium neon (632 nm), and GaAlAs or gallium-aluminum-arsenide (diode; semiconductor) (720 to 904 nm) lasers that have been used in laser Doppler flowmetry and in treating dentin hypersensitivity.[144]

In order to summarize the available data, a discussion of the application of lasers for two specific purposes will follow.

LASERS IN THE PREVENTION, DIAGNOSIS, AND TREATMENT OF CARIES

Laser irradiation of deep susceptible pits and fissures may reduce the incidence of dental caries. Once caries develops, some lasers may be effective in removing the carious lesion and sparing undemineralized dentin because of their differential absorption by water and hydroxyapatite. Furthermore, if caries exposes the pulp in young teeth, particularly those with immature apex, lasers may be able to effectively excise coronal infected pulp in pulpotomy because of their hemostatic and antibacterial properties. All these potential uses prompted a large number of investigations on the effectiveness of lasers in these applications.

Some clinicians have proposed using lasers to enhance adhesion of pit and fissure sealants; however, this was shown to not add any advantage following acid-etching—a necessity for adhesion.[177] Laser fluorescence is used by the DIAGNOdent and DIAGNOdent pen, which are laser devices that have been introduced for the diagnosis of noncavitated caries. Although these devices initially showed some promising results,[178] more recent work shows that they are best used as adjunctive devices to radiography and visual examination.[8,55,156]

Laser ablation of superficial carious lesions may be more conservative than bur preparation. A recent controlled clinical trial supports this tenet, when free-running pulsed Nd:YAG laser was used to ablate superficial pit and fissure caries in third molars scheduled for extraction (Fig. 14.22).[103] In this study, there were no histologic differences in the pulp response between the two groups.

From the perspective of the effects on the pulp, most laser applications that are employed in cutting or modifying cavities in dentin or acting directly on the pulp tissue are important. Earlier studies showed reduced permeability of dentin in vitro with an XeCl excimer laser (a laser with a relatively short wavelength of 308 nm in the ultraviolet range).[272] The apparent fusion of tubules in superficial layers of dentin was shown to occur with CO_2, Nd:YAG, and Er:YAG lasers in vitro.[314] The pulp responses to Nd:YAG and CO_2 lasers were not favorable. It was shown that Nd:YAG, and to a lesser degree CO_2 lasers, may be associated with charring and significant inflammation in the pulp compared with Er:YAG laser (Fig. 14.23).[289,314] More recently, it was reported that water cooling was necessary for laser ablation as it is for high-speed bur preparations.[45] Studies have shown that Er:YAG laser appears to induce similar responses in the pulp to those seen with high-speed bur preparations, at the level of light microscopy analysis.[78,285,286,314] However, the findings with electron microscopy were different: it was reported that while shallow cavities ablated in rat molars using Er:YAG lasers did not show changes from base-line using light microscopy and transmission electron microscopy showed disruption and degeneration of pulp peripheral nerve endings and of myelin sheath in the immediate postoperative period (Fig. 14.24).[122] This may explain the reduced sensitivity that accompanies laser cavity preparations. Thus in summary, it does not appear that the use of lasers provides

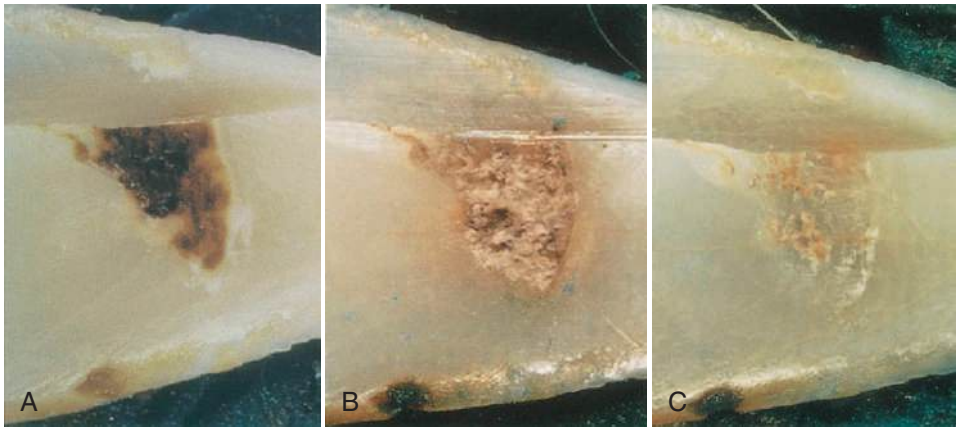


Fig. 14.22 A, Dental caries. **B**, Pulsed Nd:YAG ablation by-products (160 mJ, 10 Hz). **C**, Debris removed with acid etch and polishing. Enamel surface was faceted for reflection spectroscopy. (Reprinted with permission from Harris DM, White JM, Goodis H, et al. Selective ablation of surface enamel caries with a pulsed Nd:YAG dental laser. *Lasers Surg Med* 30:342–350, 2002.)

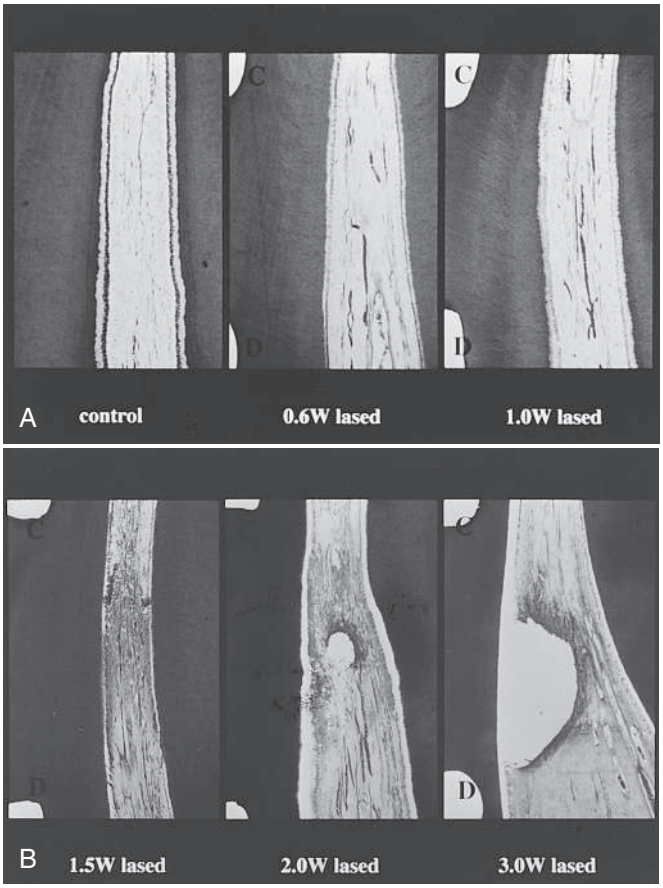


Fig. 14.23 A and **B,** Histopathologic picture of Nd:YAG laser specimen with an increasing power. There is direct relationship between the degree of pathologic changes and the increasing power of the laser. In fact, 1.5 W and the greater power cause permanent damage to the pulp. (Reprinted with permission from Kim S, Trowbridge H, Suda H. Pulpal reaction to caries and dental procedures. In: Cohen S, Burns RC, editors. *Pathways of the pulp,* ed 8, St. Louis, 2002, Mosby, pp 573–600.)

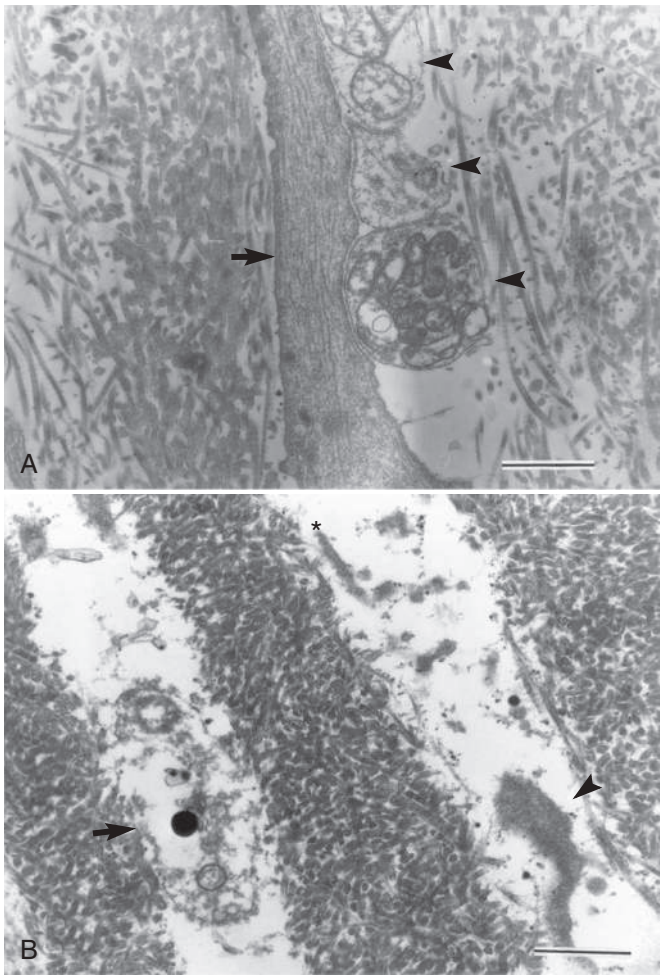


Fig. 14.24 A, Control; normal odontoblastic process *(arrow)* and a few nerve terminals *(arrowheads)* are seen in a dental tubule of rat upper first molar. **B,** Six hours after Er:YAG laser irradiation; disrupted cell membrane of a nerve terminal that contains some granular vesicles *(arrow)* and shrinkage of an odontoblastic process *(arrowhead)* are noted in dentinal tubules just under the ablated area. An asterisk indicates the irradiated side. *TEM,* ×13,700; *bar,* 1 μm. (Reprinted with permission from Inoue H, Izumi T, Ishikawa H, Watanabe K. Short-term histomorphological effects of Er:YAG laser irradiation to rat coronal dentin-pulp complex. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 97:246–250, 2004.)

predictable advantages in cavity preparation compared with traditional methods at this time.

For pulpotomy procedures, such as in primary teeth, permanent teeth with immature apex, or pulps that are exposed due to fracture and treated promptly, the lasers, particularly CO_2 lasers, may be useful in achieving precise surgical excision of coronal pulp and immediate hemostasis. A controlled clinical study showed that CO_2 laser pulpotomy was comparable to traditional methods in experimental pulpotomies of primary teeth scheduled for extraction for orthodontic reasons.[75] However, an animal study in which CO_2 and Nd:YAG lasers were compared revealed poor response with both lasers compared with calcium hydroxide.[133] An animal study using Er:YAG laser showed that the results are dependent on the power settings, in that lower energy delivered with this laser produced favorable results.[145]

LASERS IN THE TREATMENT OF DENTIN HYPERSENSITIVITY

Earlier studies have shown effectiveness ranging from 5% to 100% of low output lasers on dentin hypersensitivity.[144] One author reported a reduction of hypersensitivity in 73% of mild cases, 19% of moderate cases, and 14% of severe cases after 4 months, using a GaAlAs laser.[144] Low output lasers do not have any effects on the morphology of enamel or dentin but are thought to cause transient reduction in action potential mediated by pulp C-fibers, but not Aδ-fibers,[311] although this finding was not consistent.[214] Nd:YAG lasers have also been used in dentin hypersensitivity. Because of the higher power output, these lasers cause superficial occlusion of dentinal tubules of up to 4 μm,[173] in addition to action potential blockage within the pulp in vitro or in experimental animals.[213,214] However, a placebo-controlled clinical trial has shown that both Nd:YAG and placebo caused significant reduction in dentin hypersensitivity for up to 4 months postoperatively, but were not different from each other.[167] More recent clinical trials show that while lasers are useful for managing hypersensitivity, they do not appear to be better than less costly and more readily available alternatives.[88,301]

Pulp Reactions to Vital Bleaching Techniques

Vital bleaching techniques employ the use of strong oxidizing agents, namely, 10% carbamide peroxide and hydrogen peroxide, to bleach the enamel of teeth with vital pulp. There have been concerns about the potential for pulp irritation during these procedures because of the long amount of time during which the chemicals are in contact with the teeth, particularly if dentin with open tubules or cracks are present. Histologic or histochemical analysis of the pulp following bleaching for up to 2 weeks showed minor inflammatory changes in the pulp of the bleached teeth that were reversible.[10,92] One clinical report documented that if 16% carbamide peroxide was used, gingival irritation was evident; however, no changes in pulp vitality or in symptoms were noted. Even in patients who develop symptoms postoperatively, these tend to be reversible and can be prevented by treating the teeth with fluorides and by correcting restorative deficiencies preoperatively.[201] Clinical symptoms are likely to be due to increases in neuropeptides, such as SP, in the pulp.[46] An earlier clinical trial showed that vital bleaching using 10% carbamide peroxide in a custom tray for 6 weeks was safe for the pulp health for up to 10 years postoperatively, although the bleaching effectiveness may decline with time.[250] In one recent clinical study premolars scheduled for orthodontic extraction had 38% H_2O_2 bleaching gel with and without a halogen light source applied. There were no histologic effects on the dental pulp at 2 to 15 days after bleaching.[146] When the study was performed on incisors with the same treatment for 45 minutes, areas of coagulation necrosis could be seen in the pulp.[58] Therefore caution should be exercised when this caustic agent is used for extended bleaching.

Light activation of bleaching agents is widely used. In conjunction with a source of peroxide moieties, the light emission results in both thermal and photo catalysis of the bleaching agent, thereby releasing hydroxyl radicals and facilitating the whitening process.[169,176] The light sources can be halogen light or diode lasers, and they are reported to speed the bleaching process, allowing shade improvement over a shorter period. Some of these techniques, however, have been shown to generate sufficient heat to result in a rise in pulp tissue temperature, increasing the risk of thermal tissue damage. To this point, comparisons of bleaching with and without light activation report a higher incidence of tooth sensitivity in the light-activated bleaching group.[147] Pretreatment with a desensitizing gel of 5% nitrate potassium/2% sodium fluoride was effective in one study in reducing postoperative sensitivity after light-activated bleaching.[246]

Pulp Reactions to Periodontal Procedures

In an intact non-traumatized nontraumatized existence throughout life because it is insulated from microbial irritation in the oral cavity. Periodontal disease causes attachment loss, which exposes the root surface to the oral cavity. Occasionally, pulp inflammation secondary to severe periodontitis is observed.[161] Some reports have described bacterial infiltration through dentinal tubules of exposed root surface causing mild inflammatory changes in the pulp (Fig. 14.25).[159] However, in a case report of 25 teeth that were extracted in a patient with extensive periodontal disease, none of the pulp tissue had significant inflammatory changes.[292] It is much more common for pulp necrosis, or for failure of healing of periapical lesions, to present clinically with signs of periodontal disease than for periodontal disease to cause pulp pathosis. Primary endodontic, secondary periodontal pathosis is particularly evident if a perforation occurs in the pulp chamber or coronal third of the root during endodontic treatment and is not promptly treated, in cases of cracked teeth/vertical root fractures or in cases of congenital tooth defects such as palatal groove defects. Thus it is more likely for microbial irritants to move outward from a necrotic pulp to cause periodontal breakdown than for them to move inward from a periodontal pocket to cause irreversible pathosis in the vital pulp. The reasons for this observation are not fully understood. However, if the assumption is that bacteria may migrate through patent dentinal tubules in these situations, as is shown in Fig. 14.25, then it may be that the outward dentinal fluid flow in teeth with vital pulp contribute to the resistance to ingress of bacteria in sufficient amounts to cause a clinically significant disease process. Once the pulp degenerates, dentinal fluid flow no longer exists. Thus microbial irritants from the pulp may promote pocket formation and periodontal bone loss,[129] and the prognosis of endodontic and periodontal treatment may be related,[130] as the microbial factors may pass across dentin more readily.

Periodontal scaling and root planning may result in removal of cementum and exposure of dentin to the oral cavity. Frequently, this treatment results in dentin hypersensitivity, as discussed previously. In theory, periodontal disease and its treatment should be associated with increased incidence of pulp pathosis. In an older study, investigators induced periodontal disease in a primate model using ligatures and compared the effects on the pulp of periodontal disease with or without scaling.[24] In teeth with periodontal disease, mild chronic inflammatory changes were observed in 29% of the teeth and could be seen in areas of the pulp related to bone loss. One of 40 teeth developed pulp necrosis. In teeth that received scaling, a similar percentage of 32% developed the same mild inflammation and none developed pulp necrosis. In a later clinical study,[25] 52 periodontitis patients who had 672 teeth with vital pulp were followed up and maintained every 3 to 6 months for 4 to 13 years (mean 8.7 years). Of those teeth, 255 were bridge abutments. The results showed a significantly higher chance of pulp complications in teeth that were bridge abutments than teeth that were not abutments (15% vs. 3%; $P < .01$). Considering that both types of teeth had similar degrees of periodontal disease, the authors concluded that prosthodontic treatment is associated with pulp involvement more frequently than periodontal disease and its treatment. Another histologic analysis of 46 teeth with varying degrees of periodontal disease and coronal restorations reached a similar conclusion.[65] Furthermore, two comprehensive reviews of the topic concluded that while the potential exists for periodontal disease and its treatment to cause pulp pathosis, particularly if large lateral or accessory canals are exposed, this occurrence is rare.[102,254]

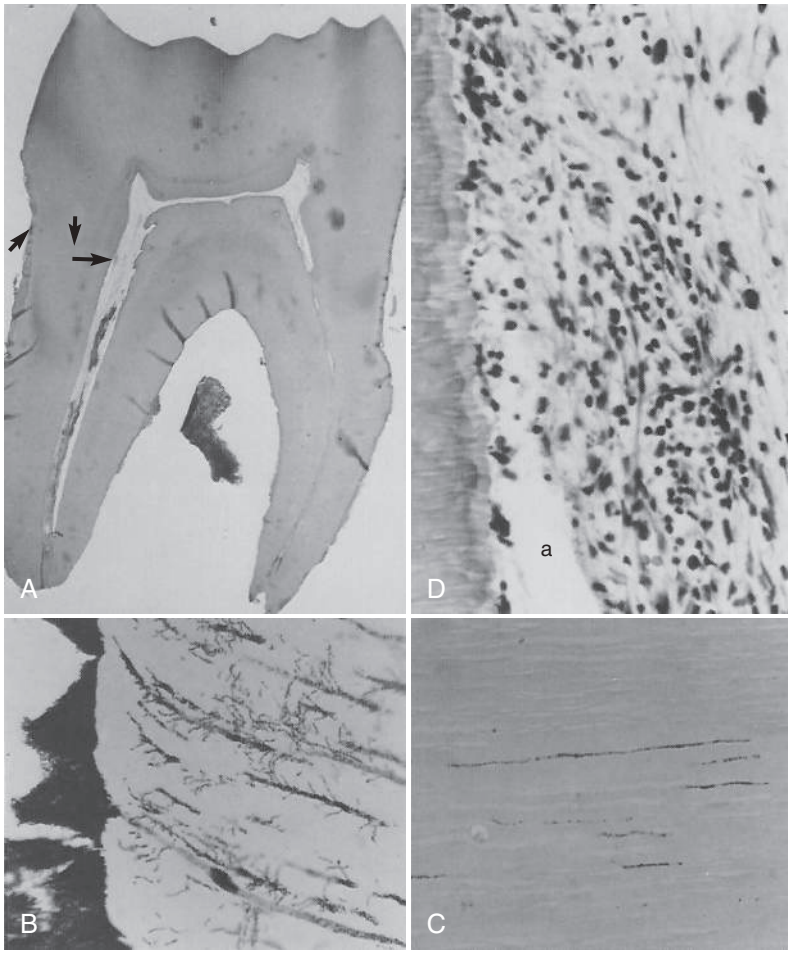


Fig. 14.25 Mandibular first molar of 49-year-old female, no pain. Clinically caries free. Calculus, periodontal disease, and bone loss from two thirds to three fourths of root. **A,** Note narrowness of pulp chamber seen throughout all serial sections (original magnification ×8.5). **B,** Bacterial plaque and adjacent dentinal tubules with bacteria (*oblique arrows* in **A**) (original magnification ×400). **C,** Farthest penetration of bacteria in dentinal tubules (*vertical arrow* in **A**) (original magnification ×400). **D,** Where dentinal tubules invaded by bacteria terminate in the pulp (*horizontal arrow* in **A**), a small but dense accumulation of lymphocytes and macrophages. *a,* Artifact; pulp tissue torn away from dentin during processing. (Original magnification ×400.) (Reprinted with permission from Langeland K. Tissue response to dental caries. *Endod Dent Traumatol* 3:149–171, 1987.)

Mechanical Irritants: Orthodontic Movement

The most conspicuous pulp change observed in response to orthodontic forces is hemodynamic. Both human and animal studies have confirmed that both lateral and intrusive forces result in an increase in PBF.[158,203,212] Furthermore, blood flow alterations are not confined to the tooth in active movement. Observed increases in blood flow are seen in teeth adjacent to the focus of movement forces, implying that directed forces on one tooth could shunt blood to proximal vessels, supplying other oral structures including teeth. If orthodontic forces are extreme, circulatory interruptions can occur, resulting in pulp necrosis.[37]

Biochemical, biologic, and histological studies of the effects of orthodontic movement have confirmed that metabolic as well as inflammatory changes can result. The dental pulp tissue respiration rate is depressed after short-term application of orthodontic force.[100] Biochemical and molecular markers confirm that apoptosis and necrosis of pulp cells are also increased subsequent to movement.[233] However, it was shown that the inflammatory mediators IL-1α and TNF-α show minor increases in the pulp during orthodontic movement, compared to the increase in periodontal tissues.[27] Histological examination of pulps in teeth subjected to intrusive forces showed vascular congestion and dilatation as well as vacuolization of the odontoblastic layer.[276–278] Most, if not all, of these effects are due to circulatory changes, and the consensus is that they are transient provided that the movement forces are not excessive.[152] It was recently shown, however, that pulp necrosis of teeth that were undergoing orthodontic treatment and had also been traumatized prior to that is significantly increased compared to teeth that had either of these conditions but not the other, particularly in lateral incisors and if the traumatized tooth had pulp canal obliteration.[19,20] As noted previously, SP appears to increase in the dental pulp as a result of acutely induced occlusal trauma.[47] Also, CGRP increases under the influence of orthodontic forces.[48] One area of growing interest with respect to orthodontic forces is their role in the pathogenesis of cervical invasive root resorption (CIRR). Studies that have categorized the potential etiology of CIRR

have documented that the history of orthodontic treatment is the most cited associated factor.[107,181] However, longitudinal observational or case control studies that show a stronger relationship between CIRR and orthodontic treatment are lacking.

Pulp Reactions to Orthodontic Surgery

It has been known for decades that osteotomies in the maxilla or mandible may cause disruption in the blood supply to teeth in the area of the surgery, with resultant inflammation and/or necrosis.[16,150,200,231] Occasionally, the teeth affected show postoperative manifestations common with traumatic injuries, such as pulp canal obliteration.[305] Animal studies have shown that if a safe distance of 5 to 10 mm is maintained between the site of the surgery and the teeth, minimal disruption occurs.[70,319] A number of studies have documented, using laser Doppler flowmetry, actual reduction in PBF immediately following maxillary Le Fort I osteotomy,[215,242,257] particularly if segmental osteotomy is performed.[76] In most cases, the blood flow is regained within months of the surgery. A modification of the Le Fort I osteotomy technique has been described in which the Le Fort I sectioning is combined with a horseshoe palatal osteotomy to spare any disruption to the descending palatine artery.[101] An examination of PBF of maxillary teeth using laser Doppler flowmetry as well as the responsiveness to electric pulp testing showed significant differences between the two surgical techniques in the postoperative recovery values (Fig. 14.26). In cases where the surgery did not disrupt the palatine artery, the PBF in the anterior teeth consistently increased without disruption in the postoperative period. A recent systematic review concluded that following a Le Fort I osteotomy, there is a decrease in pulpal vascularity and neurosensory response but only in the early postoperative period (1 to 10 days), and that this is likely a temporary effect.[243]

It should also be noted that occasionally, teeth are traumatized during endotracheal intubation for surgery that requires general anesthesia, when the surgery itself is not related to the jaws or teeth.[265]

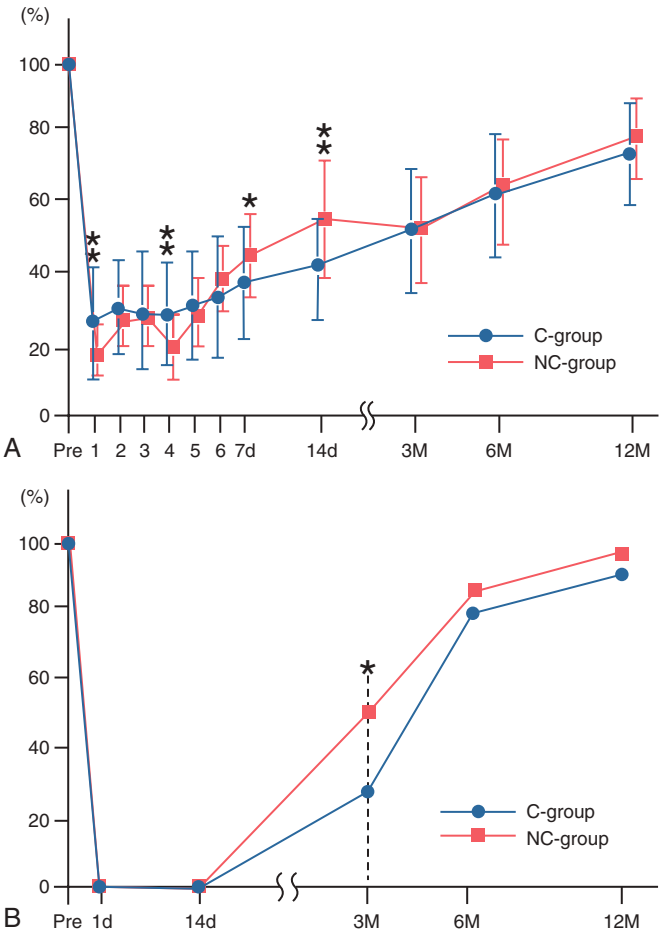


Fig. 14.26 **A,** The postoperative change of the mean pulpal blood flow in the upper incisors of the two groups. **B,** The postoperative change in the percentages of teeth (upper incisors) with positive pulpal sensibility in the two groups. *d*, Day(s) after the operation; *M*, months after the operation; *Pre*, before the operation; *bars,* SD; *, $P < .05$; **, $P < .01$. (Reprinted with permission from Harada K, Sato M, Omura K. Blood-flow and neurosensory changes in the maxillary dental pulp after differing Le Fort I osteotomies. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 97:12–17, 2004.)

Biomechanical Irritation: Parafunctional Habits

Occlusal loading of teeth effects deformation to varying degrees.[187] While enamel is largely resistant to flexure, the underlying dentin demonstrates considerable elastic and viscoelastic characteristics. As a result, defects in enamel secondary to cracks, decay, and/or restorative preparation allow cusp flexure with subsequent pulp responses, presumably due to dentinal fluid flow secondary to compression and microleakage.[110] The magnitude of the pulp response is dictated by the degree and chronicity of dentinal deformation.

Multiple factors influence the degree of tooth deformation during occlusal loading. Investigators have noted that preparation geometry has a direct impact on cuspal flexure. The width of the occlusal isthmus relative to the facio-lingual dimension of the tooth as well as the ablation of marginal ridges directly impact the degree of cusp flexure.[221,244,304] MOD preparations have been shown to cause a 50% reduction in cusp stiffness and resistance to fracture. Physical properties of the restorative material can also play a part in cusp flexure. Studies have shown that polymerization shrinkage of certain resin composites can induce an inward deflection of cusps with resultant stresses on tooth structure.[306,307]

Symptomology from cusp flexure can result from two primary sources. It has been theorized that cusp flexure results in dentin deformation, thus promoting dentinal fluid flow that activates nerve endings in the odontoblast layer of the tooth. This is supported in part by an in vitro study, which found that dentinal fluid flow could be induced by occlusal loading of restored teeth.[110] A second source of pulp pain is bacterial microleakage created by a gap at the restoration/dentin interface that is repeatedly opened during cycles of occlusal loading. If repeated cuspal flexure gives rise to a crack, dentinal exposure to bacteria and their

by-products is even greater. It is likely that in vivo, both dentinal fluid flow and bacterial access to dentinal surfaces work together to produce inflammation that is often manifested by thermal as well as biting sensitivity for the patient. In the instance of parafunctional habits, this is often combined with concussive periodontal forces to the periodontium that induce acute periradicular periodontitis, mobility, and radiographic changes.

On the cellular level, studies have shown increased levels of SP after experimentally induced occlusal trauma.[47] SP is a modulator of both sensory and immune function in the dental pulp. Alterations in physiologic levels of this mediator could result in disruptions in the pain response as well as secondary stimulation of prostaglandin E2. Furthermore, the immunostimulatory ability of neuropeptides could initiate and maintain a chronic pulp inflammation, possibly leading to necrosis. While animal studies have suggested that pulp necrosis is a possible sequela to chronic occlusal trauma, there are no controlled clinical studies that confirm this.

Dentinal cracks expose tubules that are not occluded by a smear layer and therefore offer a direct portal to the subjacent pulp. When dentinal tubules are freely exposed, there is an outward flow of dentinal fluid driven by relatively high pulp tissue pressures. Dentinal fluid is composed of proteins such as fibrinogen and serum albumin, which can coagulate and effectively block the tubule lumen, thereby limiting fluid egress and resultant dentin hypersensitivity. This phenomenon can occur within 2 days. As this serves as a short-term protective mechanism for the pulp, dentinal sclerosis and tertiary dentin formation can ultimately provide greater protection for the pulp and reduction of symptoms. Clinical interventions include the application of materials that occlude the tubules and extracoronal restorations to prevent the propagation of cracks. Cracks were also shown to harbor bacterial biofilms that can add to the irritation and inflammation of the pulp and induce periodontal pocket formation in cases with cracked teeth.[249]

Pulp Reactions to Implant Placement and Function

Osseointegrated implants are now a common option for the replacement of missing teeth. The placement of implants requires multifaceted preoperative radiographic techniques, including intraoral, tomographic, cephalometric, and panoramic imaging.[298] This ensures that implant placement fully rests in bone and does not compromise neighboring structures, including teeth. The lack of attention to the three-dimensional anatomy of the site of implant placement and the orientation of neighboring teeth may lead to the implant perforating the root and devitalizing the pulp.[180,281] The same could occur during procedures of maxillary sinus floor augmentation.[21]

It is usually recommended that implants not be placed directly at a site where a periradicular lesion, particularly one with signs of purulence, exists, as microbial irritants may interfere with osseointegration.[238] However, some data suggest that immediate implant placement in sites that have been adequately debrided is successful.[42] One systematic review corroborated this finding, provided that adequate preoperative, intraoperative, and postoperative antimicrobial measures are utilized.[52]

Case reports have also claimed that teeth with periradicular lesions may reduce the success of neighboring implants even if adequate endodontic treatment is performed.[280] To address this issue, a study was reported in which implants were placed to replace premolars in dogs and periradicular lesions were likewise induced, some of which were treated nonsurgically or with surgical and nonsurgical treatment.[263] The results showed that the presence of treated or untreated periradicular lesions did not affect the long-term osseointegration of implants that were already osseointegrated. Clinical cases with complete resolution of periapical lesions that also involve neighboring implants, following adequate endodontic treatment, have been published.[170,256,275]

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